

Lecture 12: The Basis (continued)

Instructors:

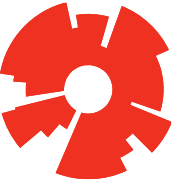
Christoph Lohrmann (christophl@ru.is)

[Based on slides for „Decoding the Matrix“]



WEEK

SIX



Agenda

Chapter 5

Lecture 12:

Linear Dependence - Hereditary
Linear Dependence - Importance
Basis
Unique Representation via Basis
Change of Basis
Simplified Exchange Lemma
Exchange Lemma





Lecture 11 - Summary

Concepts

Basis

Theorems & Lemmas

Linear-Dependence
Lemma

Superfluous-Vector
Lemma

Minimum
Spanning Forest

Tools

Grow
Algorithm

Shrink
Algorithm

Span

Linear
Combination

Coefficients

Previous

Vector Space

Coordinate
System

Coordinate
Representation

Linear
Independence





Recap - Linear Dependence

Superfluous-Vector Lemma: For any set S and any vector $\mathbf{v} \in S$, if $\mathbf{v}$ can be written as a linear combination of the other vectors in S then $\text{Span}(S - \{\mathbf{v}\}) = \text{Span } S$

- Is there a linear combinations of the vectors (excl. v) that can reach v ?
- Solve it via a **(general) system of linear equations**

Definition: Vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are *linearly dependent* if the zero vector can be written as a **nontrivial** linear combination of the vectors:

$$\mathbf{0} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$$

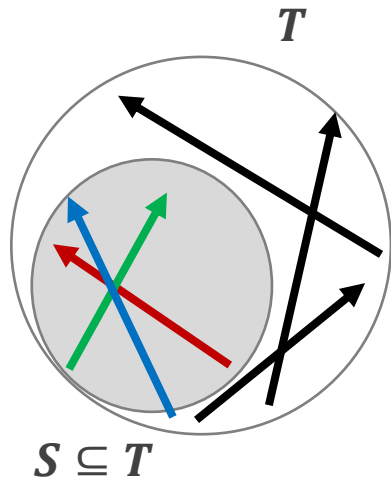
- Is there only the **trivial** linear combinations of the vectors to reach the zero vector?
- Solve it via a **homogeneous system of linear equations**



Linear Dependence - Hereditary (I)

Lemma: If a finite set S of vectors is linearly dependent and S is a subset of T then T is linearly dependent.

In graphs, if a set S of edges includes a cycle then a superset of S also includes a cycle.



If S is linearly **dependent**, so is the superset T that contains it

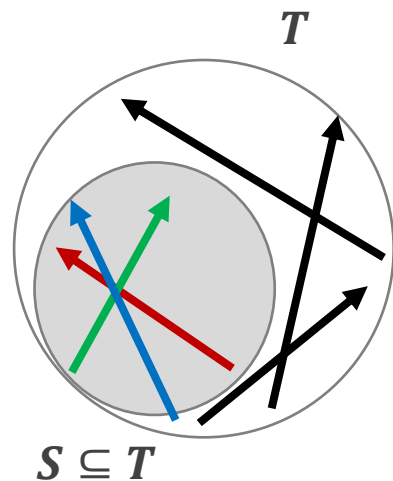
$$\begin{array}{ccc}
 \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \right\} & \xrightarrow{\quad} & \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \\
 S & & T \\
 \text{(linear} & & \text{(linear} \\
 \text{dependent)} & & \text{dependent)}
 \end{array}$$



Linear Dependence - Hereditary (II)

Lemma: If a finite set S of vectors is linearly dependent and S is a subset of T then T is linearly dependent.

Obviously not true about linear independence



If S is linearly **independent**,
superset T that contains it may not
be linearly independent as well

$$\begin{array}{cc} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} & \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right\} \\ S & T \\ \text{(linear independent)} & \text{(linear dependent)} \end{array}$$

$$\begin{array}{cc} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} & \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \\ S & T \\ \text{(linear independent)} & \text{(linear independent)} \end{array}$$



Linear Dependence - Hereditary (III)



Proof: If the zero vector can be written as a nontrivial linear combination of some vectors, it can be so written even if we allow some extra vectors to be in the linear combination because we can use zero coefficients on the extra vectors.

More formal proof: Write $S = \{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ and $T = \{\mathbf{s}_1, \dots, \mathbf{s}_n, \mathbf{t}_1, \dots, \mathbf{t}_k\}$. Suppose S is linearly dependent. Then there are coefficients $\alpha_1, \dots, \alpha_n$, not all zero, such that

$$\mathbf{0} = \alpha_1 \mathbf{s}_1 + \dots + \alpha_n \mathbf{s}_n$$

Therefore

$$\mathbf{0} = \alpha_1 \mathbf{s}_1 + \dots + \alpha_n \mathbf{s}_n + 0 \mathbf{t}_1 + \dots + 0 \mathbf{t}_k$$

Adding vectors can
not undo the linear
dependence

which shows that the zero vector can be written as a nontrivial linear combination of the vectors of T , i.e. that T is linearly dependent.

QED



Linear Dependence - Importance (I)

How can we tell if vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly dependent?

By linear-combinations definition, $\mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly dependent iff there is a

nonzero vector $\begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$ such that $\begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} = \mathbf{0}$

Therefore, $\mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly dependent iff the null space of the matrix is nontrivial.

= How can we tell if the null space of a matrix is trivial?

Instead of looking at them as vectors, we look at them as a matrix (containing these vectors)



Linear Dependence - Importance (II)

How can we tell if vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly dependent?

= How can we tell if the null space of a matrix is trivial?

Recall: *solution set of a homogeneous linear system*

$$\begin{array}{rcl} \mathbf{a}_1 \cdot \mathbf{x} & = & 0 \\ & \vdots & \\ \mathbf{a}_m \cdot \mathbf{x} & = & 0 \end{array} \quad \text{is the null space of matrix} \quad \left[\begin{array}{c} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{array} \right].$$

= How can we tell if the solution set of a homogeneous linear system is trivial?

**Linearly
independent vectors**



**Trivial
Nullspace of
Matrix**

**Trivial
Solution Set
of homog. LS**



Linear Dependence - Importance (III)

Recall:

If $\mathbf{u}_1$ is a solution to a linear system $\mathbf{a}_1 \cdot \mathbf{x} = \beta_1, \dots, \mathbf{a}_m \cdot \mathbf{x} = \beta_m$ then

$$\{\text{solutions to linear system}\} = \{\mathbf{u}_1 + \mathbf{v} : \mathbf{v} \in \mathcal{V}\}$$

where $\mathcal{V} = \{\text{solutions to corresponding homogeneous linear system}$

$$\mathbf{a}_1 \cdot \mathbf{x} = 0, \dots, \mathbf{a}_m \cdot \mathbf{x} = 0\}$$

Thus the question is the same as:

= How can we tell if a solution $\mathbf{u}_1$ to a linear system is the *only* solution?

Linear Dependence - Importance (IV)



The question

How can we tell if vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly dependent?

Vectors

is the same as:

How can we tell if the null space of a matrix is trivial?

Matrix

is the same as:

How can we tell if the solution set of a homogeneous linear system is trivial?

Homogeneous LS

is the same as:

How can we tell if a solution $\mathbf{u}_1$ to a linear system is the only solution?

General LS



Basis (I)

If they successfully finish, the Grow algorithm and the Shrink algorithm each find a set of vectors spanning the vector space $\mathcal{V}$. In each case, the set of vectors found is linearly independent.

Definition: Let $\mathcal{V}$ be a vector space. A *basis* for $\mathcal{V}$ is a linearly independent set of generators for $\mathcal{V}$.



Thus a set S of vectors of $\mathcal{V}$ is a *basis* for $\mathcal{V}$ if S satisfies two properties:

Property B1 (*Spanning*) $\text{Span } S = \mathcal{V}$, and

Property B2 (*Independent*) S is linearly independent.

Most important definition in linear algebra.





Basis (II)

Definition: Let $\mathcal{V}$ be a vector space. A *basis* for $\mathcal{V}$ is a linearly independent set of generators for $\mathcal{V}$.



Mr. guilty is innocent! What is the **basis** for your accusations?

$[1, 0, 0]$, $[0, 1, 0]$ and $[0, 0, 1]$

Damn! ok, you win





Remember (last slide)

Property B1 (*Spanning*)Property B2 (*Independent*)

Basis (III)



Let $\mathcal{V} = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} \right\}$. Is the set of vectors $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} \right\}$ are basis for $\mathcal{V}$?

B1 Since the vectors used to define the span for $\mathcal{V}$ are the same as in the set of vectors S , it is obvious that the vectors in S span $\mathcal{V}$.



That was easy!

B2 Check if the zero vector can be obtained with a nontrivial linear combination of the vectors in S . After **solving the homogeneous system of linear equations**, we would see that (e.g.,):

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$$

Nonzero coefficients
→ B2 violated!

The set *is* spanning but is *not* independent so not a basis



Basis (IV)

(Subset of the
original vectors)

Let's see if e.g., just the first two vectors $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$ are a basis for $\mathcal{V} = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} \right\}$

B1

Recall that we determined last vector that e.g., $\begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$ is a **superfluous vector**, as it is in $\text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$.

After having determined that it is a superfluous vector, we know that

$$\text{Span}\{S\} = \text{Span}\{S - \{v\}\}$$

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

→ Spanning $\mathcal{V}$



Basis (V)

B2

$$\mathbf{0} = \alpha_1 v_1 + \alpha_2 v_2$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha_1 + 2\alpha_2 \\ 2\alpha_1 + \alpha_2 \\ 3\alpha_1 \end{bmatrix}$$

System of **homog. LEs**

$$\begin{cases} \alpha_1 + 2\alpha_2 = 0 \\ 2\alpha_1 + \alpha_2 = 0 \\ 3\alpha_1 = 0 \end{cases}$$



$$\begin{cases} (\mathbf{0}) + 2\alpha_2 = 0 \\ 2(\mathbf{0}) + \alpha_2 = 0 \\ \alpha_1 = \mathbf{0} \end{cases} \iff \begin{cases} \alpha_2 = 0 \\ \alpha_2 = 0 \\ \alpha_1 = 0 \end{cases}$$

→ **Linearly independent**
(only trivial linear combination)

Since the two vectors are **spanning** $\mathcal{V}$ and are **linearly independent**, they form a basis for $\mathcal{V}$





Basis (VI)

Example: A simple basis for $\mathbb{R}^3$: the standard generators $\mathbf{e}_1 = [1, 0, 0]^T$, $\mathbf{e}_2 = [0, 1, 0]^T$, $\mathbf{e}_3 = [0, 0, 1]^T$.

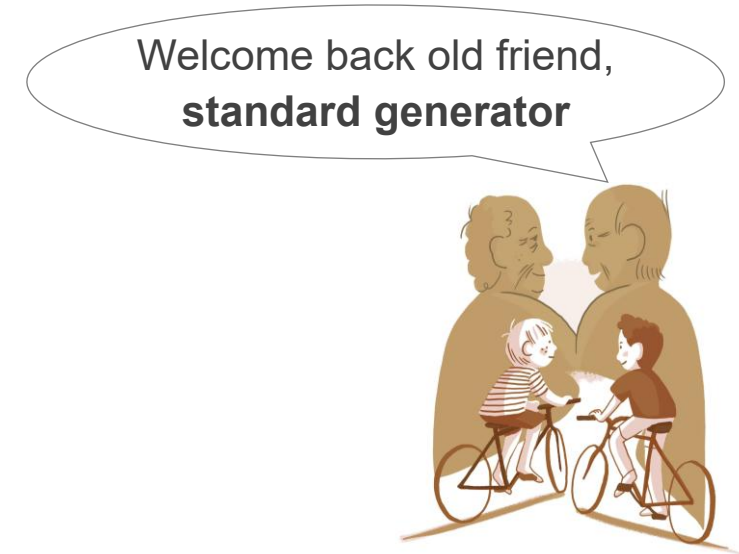
B1 ▶ *Spanning:* For any vector $[x, y, z] \in \mathbb{R}^3$,

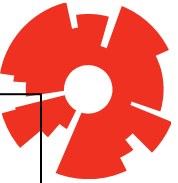
$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

B2 $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \longrightarrow \begin{cases} \alpha_1 = 0 \\ \alpha_2 = 0 \\ \alpha_3 = 0 \end{cases}$ **System of homog. LEs**

Then $\alpha_1 = \alpha_2 = \alpha_3 = 0$.

Instead of “standard generators”, we call them standard basis vectors. We refer to $\{[1, 0, 0], [0, 1, 0], [0, 0, 1]\}$ as standard basis for $\mathbb{R}^3$.





Basis (VII)

Note: Pay attention to the vector space we are checking. Before: the span of the same given vectors, now $\mathbb{R}^3$ (or a different set of vectors)

Is the set of vectors $S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ are basis for $\mathbb{R}^3$?

B1 Since we know the three standard generators span $\mathbb{R}^3$, we can check if these standard generators are in the span of the given vectors.

$$e_1: \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_1 + \alpha_2 \\ \alpha_1 + \alpha_2 + \alpha_3 \\ \alpha_1 + \alpha_3 \end{bmatrix} \longrightarrow \begin{cases} \alpha_1 = 1 \\ \alpha_2 = 0 \\ \alpha_3 = -1 \end{cases} \quad (\text{some intermediate steps skipped})$$

$$e_2: \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_1 + \alpha_2 \\ \alpha_1 + \alpha_2 + \alpha_3 \\ \alpha_1 + \alpha_3 \end{bmatrix} \longrightarrow \begin{cases} \alpha_1 = -1 \\ \alpha_2 = 1 \\ \alpha_3 = 1 \end{cases} \quad (\text{some intermediate steps skipped})$$

$$e_3: \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_1 + \alpha_2 \\ \alpha_1 + \alpha_2 + \alpha_3 \\ \alpha_1 + \alpha_3 \end{bmatrix} \longrightarrow \begin{cases} \alpha_1 = 1 \\ \alpha_2 = -1 \\ \alpha_3 = 0 \end{cases} \quad (\text{some intermediate steps skipped})$$

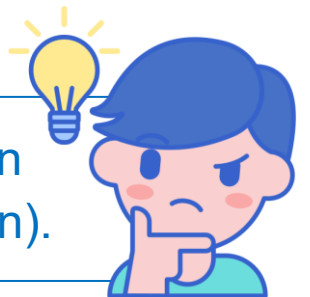


Basis (VIII)

B1 (continued from previous slide)

Since $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ can be written in terms of these new vectors, every vector in $\text{Span} \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is in span of new vectors. Thus $\mathbb{R}^3$ equals span of new vectors.

If we can “reach” the **standard generators** using them, we can “reach” anything that the standard generators can “reach” (Span).





Basis (IX)

System of **homog.** LEs

B2

$$\mathbf{0} = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha_1 + \alpha_2 \\ \alpha_1 + \alpha_2 + \alpha_3 \\ \alpha_1 + \alpha_3 \end{bmatrix}$$

$$\begin{cases} \alpha_1 + \alpha_2 = 0 \\ \alpha_1 + \alpha_2 + \alpha_3 = 0 \\ \alpha_1 + \alpha_3 = 0 \end{cases}$$



$$\begin{cases} \alpha_1 + \alpha_2 = 0 \\ \alpha_3 = \mathbf{0} \\ \alpha_1 + (\mathbf{0}) = 0 \end{cases}$$



$$\begin{cases} (\mathbf{0}) + \alpha_2 = 0 \\ \alpha_3 = 0 \\ \alpha_1 = 0 \end{cases}$$



$$\begin{cases} \alpha_2 = 0 \\ \alpha_3 = 0 \\ \alpha_1 = 0 \end{cases}$$

→ **Linearly independent**
(only trivial linear combination)

Since the two vectors are **spanning** $\mathbb{R}^3$ and are **linearly independent**, they form a basis for $\mathbb{R}^3$





Unique Representation via Basis (I)

- We represent vector $\mathbf{v}$ by its *coordinate representation* $[\alpha_1, \dots, \alpha_n]$

vector (being represented)



$$\mathbf{v} = \mathbf{A}\mathbf{u} = \alpha_1 \mathbf{a}_1 + \dots + \alpha_n \mathbf{a}_n$$

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \dots & \mathbf{a}_n \end{bmatrix} \quad \mathbf{u} = \begin{bmatrix} \alpha_1 \\ \dots \\ \alpha_n \end{bmatrix}$$



Do you think I can
uniquely represent $\mathbf{v}$?
Maybe there is another
representation?



Unique Representation via Basis (II)



E.g., previously, we said that the set of vectors $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} \right\}$ is **not a basis** for (our given) $\mathcal{V}$.

Is the **representation unique**? Let's see e.g., with $\begin{bmatrix} -3 \\ -3 \\ -3 \end{bmatrix}$

$$\begin{bmatrix} -3 \\ -3 \\ -3 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} \alpha_1 + 2\alpha_2 + 3\alpha_3 \\ 2\alpha_1 + \alpha_2 + 3\alpha_3 \\ 3\alpha_1 + 3\alpha_3 \end{bmatrix} \longrightarrow \begin{cases} \alpha_1 + 2\alpha_2 + 3\alpha_3 = -3 \\ 2\alpha_1 + \alpha_2 + 3\alpha_3 = -3 \\ 3\alpha_1 + 3\alpha_3 = -3 \end{cases}$$

(reduce number of unknowns in second equation)

$$\longleftrightarrow \dots \longleftrightarrow \begin{cases} \alpha_2 = \alpha_1 \\ \alpha_3 = -1 - \alpha_1 \\ 0 = 0 \end{cases}$$

(some steps not shown)



Unique Representation via Basis (III)

$$\begin{bmatrix} -3 \\ -3 \\ -3 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} \alpha_1 + 2\alpha_2 + 3\alpha_3 \\ 2\alpha_1 + \alpha_2 + 3\alpha_3 \\ 3\alpha_1 + 3\alpha_3 \end{bmatrix} \longrightarrow \begin{cases} \alpha_1 + 2\alpha_2 + 3\alpha_3 = -3 \\ 2\alpha_1 + \alpha_2 + 3\alpha_3 = -3 \\ 3\alpha_1 + 3\alpha_3 = -3 \end{cases}$$

(reduce number of unknowns in second equation)

$$\longleftrightarrow \dots \longleftrightarrow \begin{cases} \alpha_2 = \alpha_1 \\ \alpha_3 = -1 - \alpha_1 \\ 0 = 0 \end{cases}$$

(some steps not shown)

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 \\ -1 - \alpha_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} + \alpha_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Not a unique representation – for any α_1 , there is a different representation of point $\begin{bmatrix} -3 \\ -3 \\ -3 \end{bmatrix}$



Unique Representation via Basis (IV)

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 \\ -1 - \alpha_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} + \alpha_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Not a unique representation – for any α_1 , there is a different representation of point $\begin{bmatrix} -3 \\ -3 \\ -3 \end{bmatrix}$

e.g. $\alpha_1 = 0 \rightarrow \mathbf{u} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$

coordinates

$$A\mathbf{u} = 0 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - 1 \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ -3 \\ -3 \end{bmatrix}$$

e.g. $\alpha_1 = 1 \rightarrow \mathbf{u} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

coordinates

$$A\mathbf{u} = 1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ -3 \\ -3 \end{bmatrix}$$

More than **one** representation $\rightarrow$ not unique representation



Unique Representation via Basis (V)

Question: How can we ensure that each point has only one coordinate representation?

Answer: The generators $\mathbf{a}_1, \dots, \mathbf{a}_n$ should form a basis.



Unique-Representation Lemma Let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be a basis for $\mathcal{V}$. For any vector $\mathbf{v} \in \mathcal{V}$, there is exactly one representation of $\mathbf{v}$ in terms of the basis vectors.



Unique Representation via Basis (VI)



Unique-Representation Lemma Let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be a basis for $\mathcal{V}$. For any vector $\mathbf{v} \in \mathcal{V}$, there is exactly one representation of $\mathbf{v}$ in terms of the basis vectors.

$$A = \begin{bmatrix} \mathbf{a}_1 & \dots & \mathbf{a}_n \end{bmatrix} \quad u = \begin{bmatrix} \alpha_1 \\ \dots \\ \alpha_n \end{bmatrix}$$

Don't worry, I'm a **basis** so you are a **unique representation** of v using my help!

Do you think I can **uniquely represent** v ?
Maybe there is another representation?





Unique Representation via Basis (VII)



Proof: Let $\mathbf{v}$ be any vector in $\mathcal{V}$.

The vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$ span $\mathcal{V}$, so there is at least one representation of $\mathbf{v}$ in terms of the basis vectors.

Suppose there are two such representations: (using the same vectors but different coefficients)

$$\mathbf{v} = \alpha_1 \mathbf{a}_1 + \dots + \alpha_n \mathbf{a}_n = \beta_1 \mathbf{a}_1 + \dots + \beta_n \mathbf{a}_n$$

We get the zero vector by subtracting one from the other:

$$\begin{aligned} \mathbf{0} &= \alpha_1 \mathbf{a}_1 + \dots + \alpha_n \mathbf{a}_n - (\beta_1 \mathbf{a}_1 + \dots + \beta_n \mathbf{a}_n) \\ &= (\alpha_1 - \beta_1) \mathbf{a}_1 + \dots + (\alpha_n - \beta_n) \mathbf{a}_n \end{aligned}$$

Since the vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$ are linearly independent, the coefficients $\alpha_1 - \beta_1, \dots, \alpha_n - \beta_n$ must all be zero, so the two representations are really the same. QED



Representation in terms of a Basis (I)

Suppose we have a basis $\mathbf{a}_1, \dots, \mathbf{a}_n$ for some vector space $\mathcal{V}$.

Function
 $f : \mathbb{F}^n \longrightarrow \mathcal{V}$ defined by $f(\mathbf{x}) = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} \mathbf{x} \end{bmatrix}$ is



- ▶ *onto* (because $\mathbf{a}_1, \dots, \mathbf{a}_n$ are generators for $\mathcal{V}$) (B1: Spanning)
- ▶ *one-to-one* (by Unique-Representation Lemma)

so f is an invertible function. and matrix A that implements f is also invertible



Representation in terms of a Basis (II)

(coordinate representation of b in terms of the columns of A (basis))

$$\begin{array}{c} \text{(basis)} \end{array} \quad \begin{array}{c} \text{Coordinates} \end{array} \quad \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} \mathbf{u} \end{bmatrix} = \begin{bmatrix} \mathbf{b} \end{bmatrix} \quad \text{Original Vector (was represented by } \mathbf{u} \text{)}$$

How to get from the coordinate representation u to the original vector represented (in terms of the standard basis)?

Multiplying by basis $a_1, \dots, a_n$ brings us “back” (the inverse would bring us to u)

Just multiply u with A as $Au = b$



That was easy!





Change of Basis (I)

Now suppose $\overset{\text{(basis)}}{\mathbf{a}_1, \dots, \mathbf{a}_n}$ is one basis for $\mathcal{V}$ and $\overset{\text{(basis)}}{\mathbf{c}_1, \dots, \mathbf{c}_k}$ is another.

$$\text{Define } f(\mathbf{x}) = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} \mathbf{x} \end{bmatrix} \text{ and define } g(\mathbf{y}) = \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k \end{bmatrix} \begin{bmatrix} \mathbf{y} \end{bmatrix}.$$

Then both f and g are invertible functions.

Assume we want to change from the **red basis** to the **blue basis**
for some original vector



Change of Basis (II)

The function $g^{-1} \circ f$ maps

- ▶ from coordinate representation of a vector in terms of $\mathbf{a}_1, \dots, \mathbf{a}_n$ (basis)
- ▶ to coordinate representation of a vector in terms of $\mathbf{c}_1, \dots, \mathbf{c}_k$ (basis)



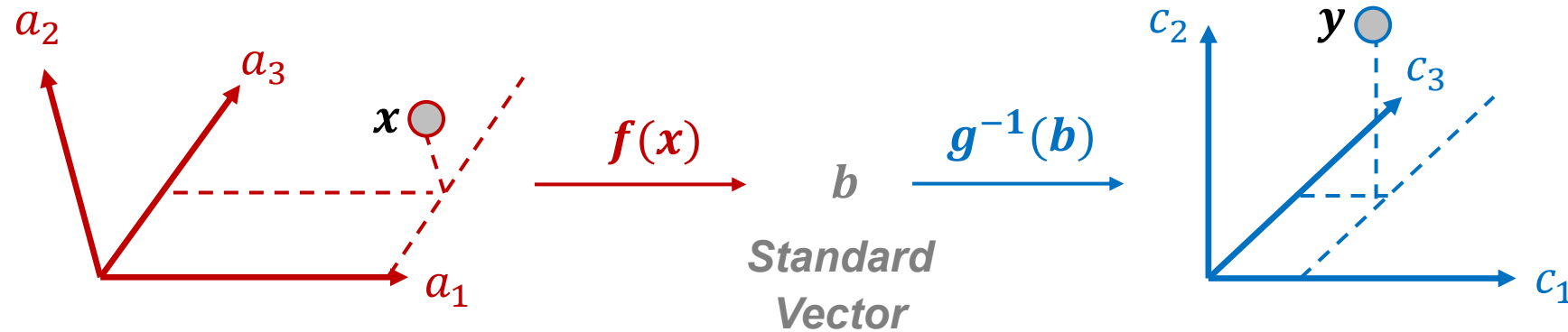
Two Steps: (1) get back from **first basis** representation to original vector, then
(2) Represent original vector in terms of the **other basis**



Change of Basis (III)

x is the coordinate representation of b using basis $a_1, \dots, a_n$

y is the coordinate representation of b using basis $c_1, \dots, c_k$



$$g^{-1}(f(x)) = \underbrace{\begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k \end{bmatrix}^{-1}}_{C^{-1}} \underbrace{\begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}}_A \begin{bmatrix} \mathbf{x} \end{bmatrix}$$



Change of Basis (IV)

Proposition: If $\mathbf{a}_1, \dots, \mathbf{a}_n$ and $\mathbf{c}_1, \dots, \mathbf{c}_k$ are bases for $\mathbb{F}^m$ then multiplication by the matrix

$$B = \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_k \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}$$

We simply give the matrix that does $g^{-1}(f(x))$ a name

maps

- ▶ from coordinate representation of a vector in terms of $\mathbf{a}_1, \dots, \mathbf{a}_n$ (basis)
- ▶ to coordinate representation of a vector in terms of $\mathbf{c}_1, \dots, \mathbf{c}_k$ (basis)

Conclusion: Given two bases of $\mathbb{F}^m$, there is a matrix B such that multiplication by B converts from one coordinate representation to the other.



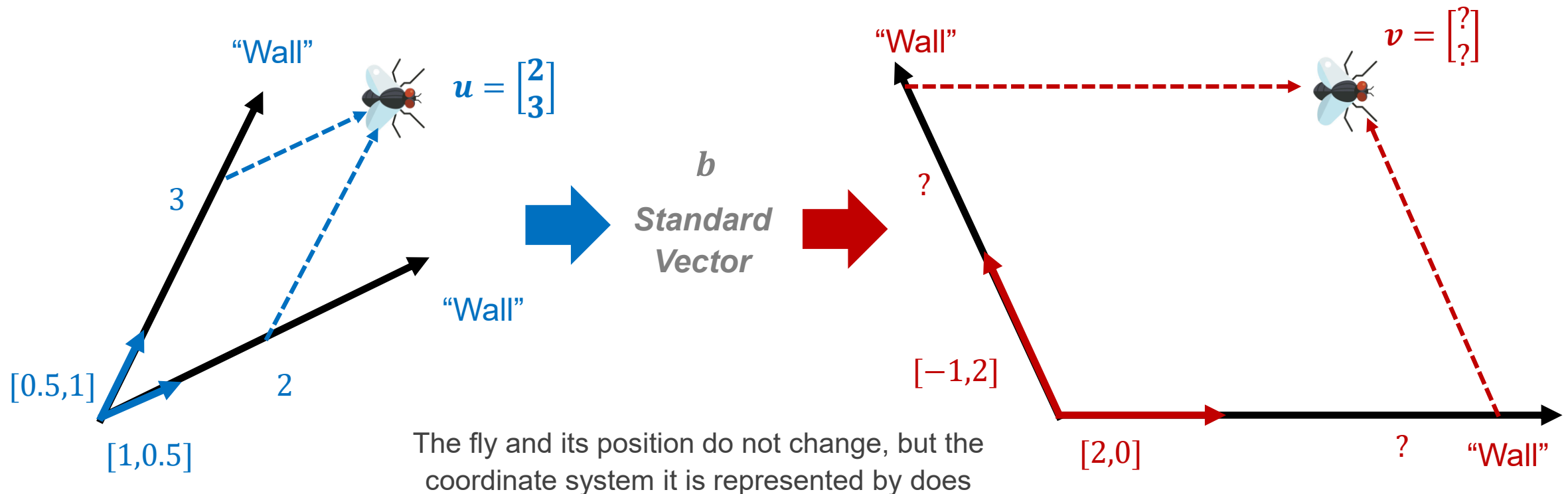
I have to find a new set of walls to live in



Change of Basis (VIII)

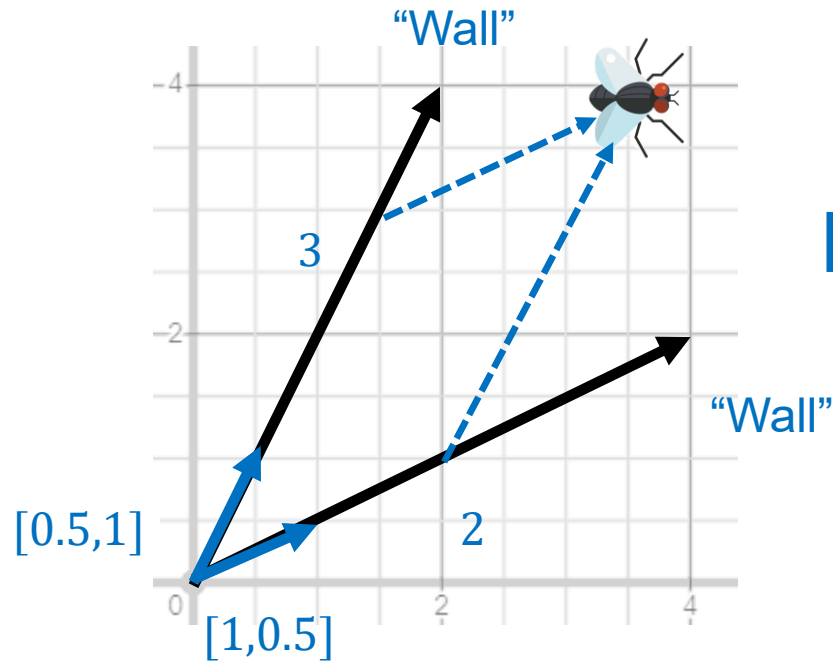
Example: Map from the coordinate representation with respect to the vectors $\left\{ \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}, \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \right\}$ (basis) to the coordinate representation with respect to the vectors $\left\{ \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$ (basis).

What are the new coordinates? What is the matrix that performs the change of basis?





Change of Basis (IX)



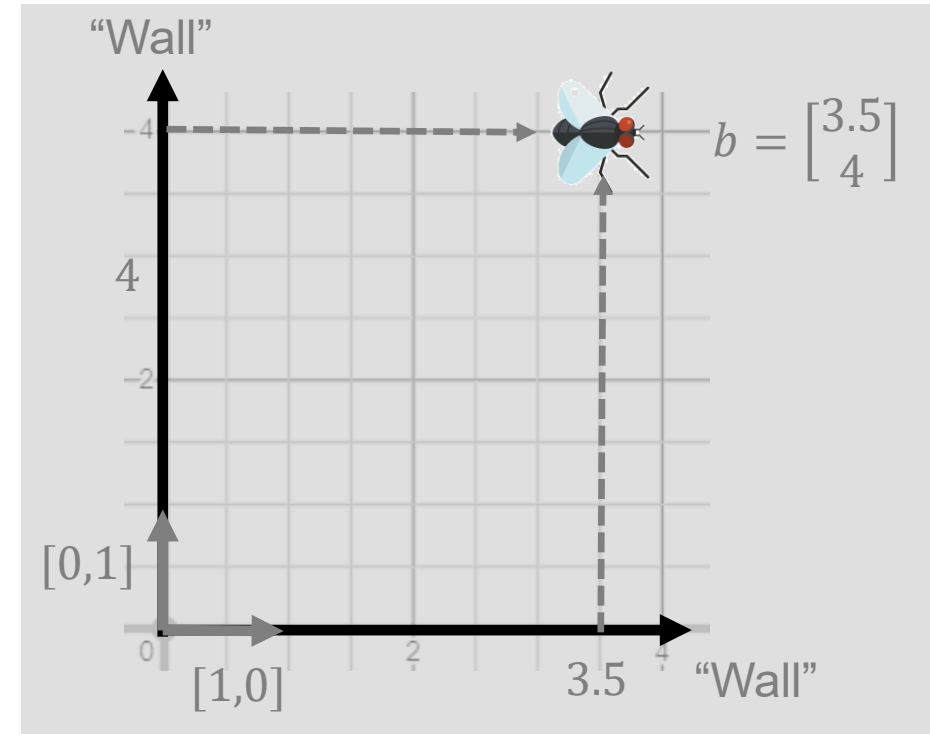
b
Standard
Vector

$$A = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

axes (columns)

$$u = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

coordinates

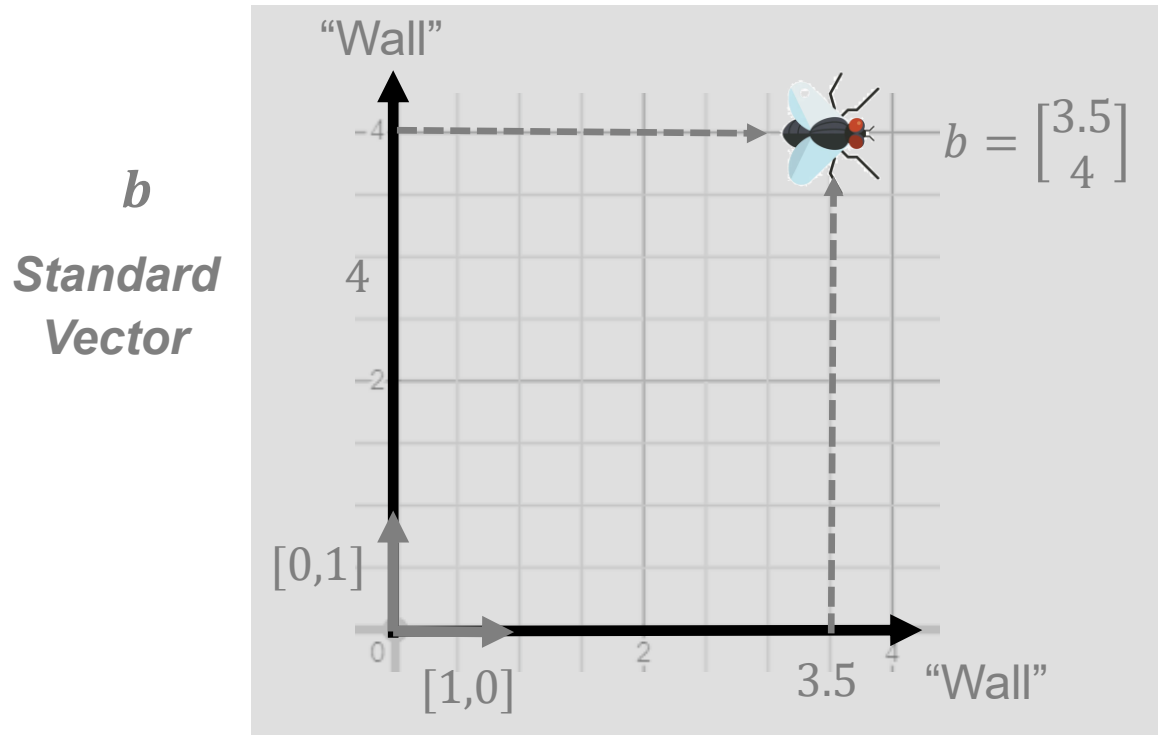


$$b = Au = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} * \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix}$$

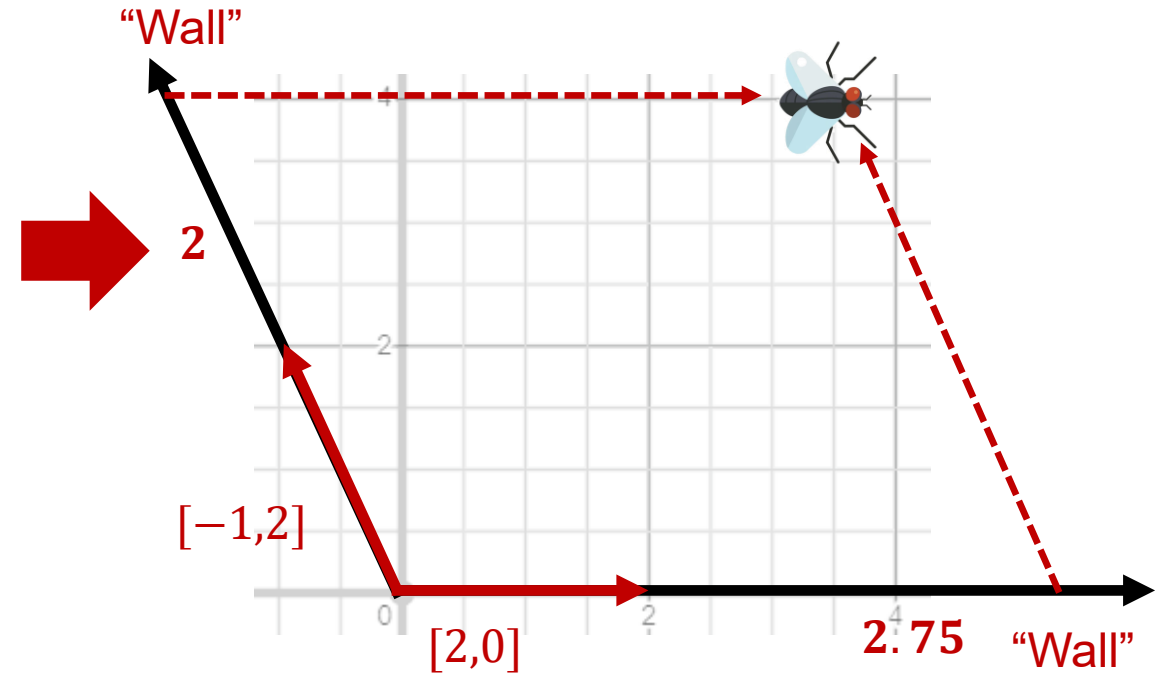
(Inverse given here – no need to calculate it)



Change of Basis (X)



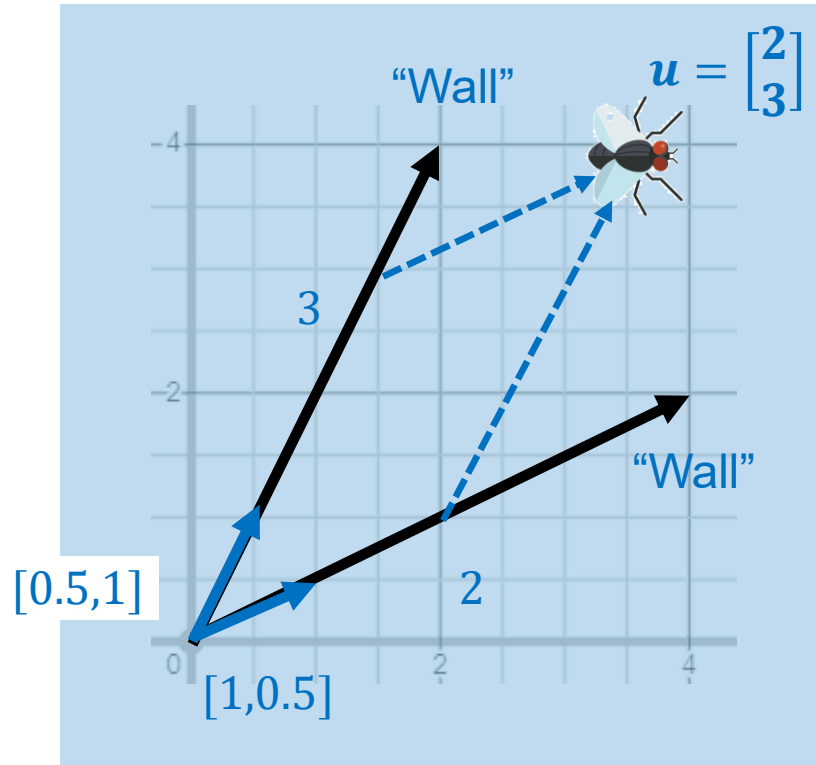
$$C^{-1}b = v$$



$$C^{-1}b = \begin{bmatrix} 0.5 & 0.25 \\ 0 & 0.5 \end{bmatrix} * \begin{bmatrix} 3.5 \\ 4 \end{bmatrix} = \begin{bmatrix} 2.75 \\ 2 \end{bmatrix}$$

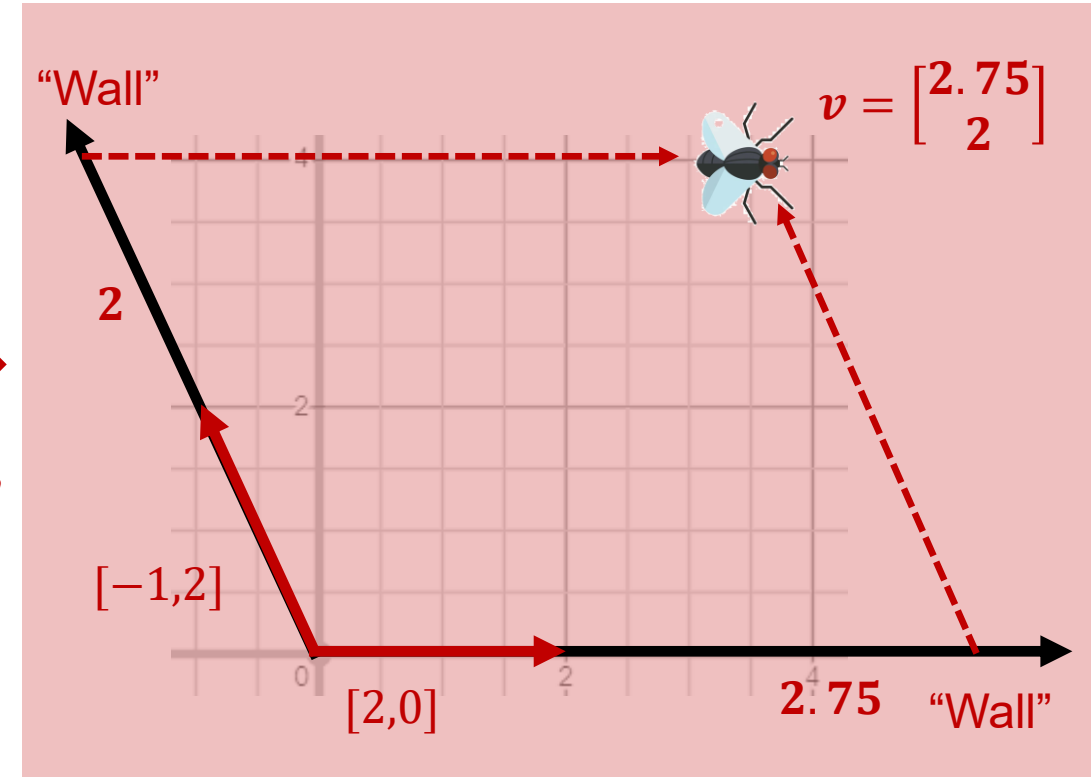


Change of Basis (XI)



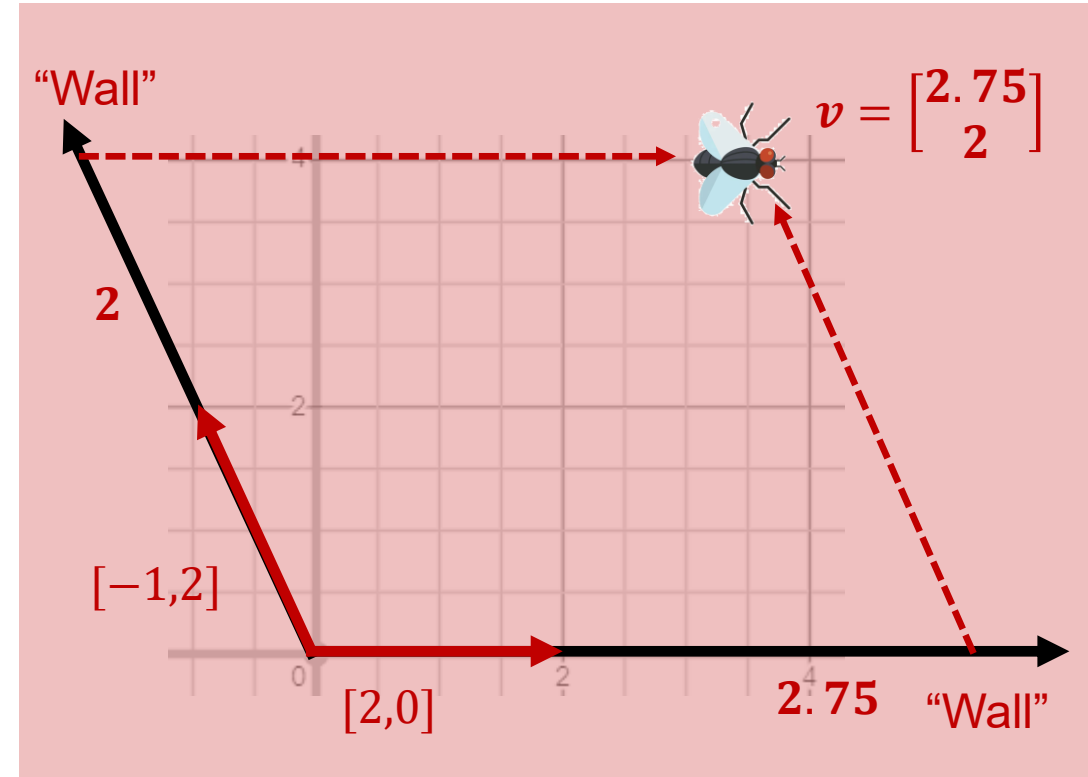
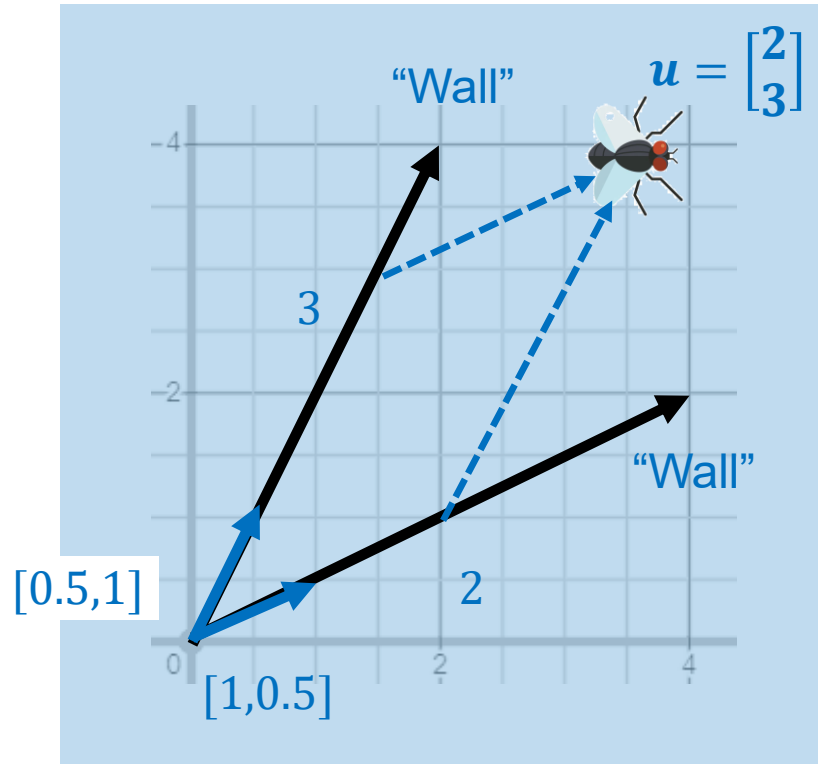
$b = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix}$
 $b = Au$

Standard Vector
 $C^{-1}b = v$





Change of Basis (XII)



$$B = C^{-1}A = \begin{bmatrix} 0.5 & 0.25 \\ 0 & 0.5 \end{bmatrix} * \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 0.625 & 0.5 \\ 0.25 & 0.5 \end{bmatrix}$$

Interpretation: $v = Bu$, which using matrix B converts coordinates expressed in the **blue basis** to coordinates in the **red basis**

Simplified Exchange Lemma (I)

We need a tool to iteratively transform one set of generators into another.

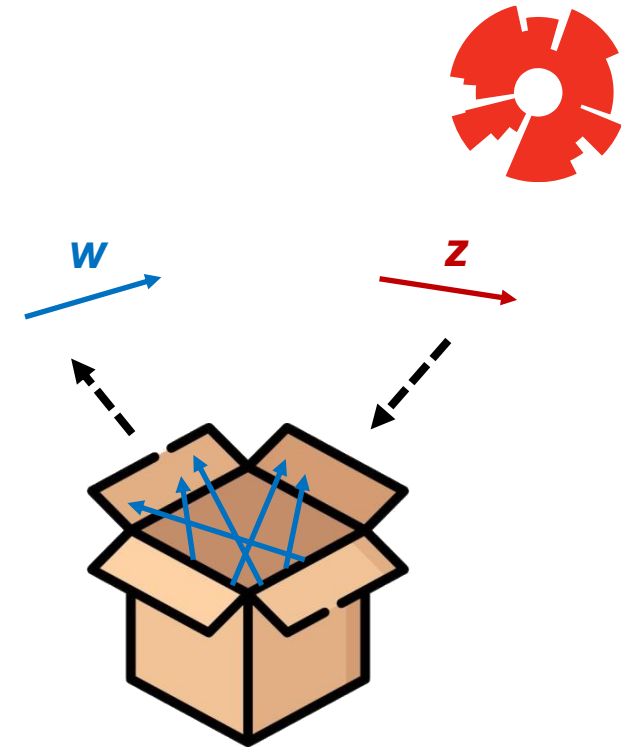
- ▶ You have a set S of vectors.
- ▶ You have a vector z you want to inject into S .
- ▶ You want to maintain same size so must eject a vector from S .
- ▶ You want the span to not change.

Exchange Lemma tells you how to choose vector to eject.

Simplified Exchange Lemma:

- ▶ Suppose S is a set of vectors.
- ▶ Suppose $\mathbf{z}$ is a nonzero vector in $\text{Span } S$.
- ▶ Then there is a vector $\mathbf{w}$ in S such that

$$\text{Span } (S \cup \{\mathbf{z}\} - \{\mathbf{w}\}) = \text{Span } S$$



... eventually ...





Simplified Exchange Lemma (II)

Simplified Exchange Lemma: Suppose S is a set of vectors, and $\mathbf{z}$ is a nonzero vector in $\text{Span } S$. Then there is a vector $\mathbf{w}$ in S such that $\text{Span } (S \cup \{\mathbf{z}\} - \{\mathbf{w}\}) = \text{Span } S$.

Proof: Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Since $\mathbf{z}$ is in $\text{Span } S$, can write

$$\mathbf{z} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$$

We put one vector in ...

By Superfluous-Vector Lemma, $\text{Span } (S \cup \{\mathbf{z}\}) = \text{Span } S$.

Since $\mathbf{z}$ is nonzero, at least one of the coefficients is nonzero, say α_i .

Rewrite as

$$\mathbf{z} - \alpha_1 \mathbf{v}_1 - \dots - \alpha_{i-1} \mathbf{v}_{i-1} - \alpha_{i+1} \mathbf{v}_{i+1} - \dots - \alpha_n \mathbf{v}_n = \alpha_i \mathbf{v}_i$$

$\mathbf{w}$ vector can be expressed as a linear combination of the remaining vectors in S and $\mathbf{z}$

Divide through by α_i :

$$(1/\alpha_i)\mathbf{z} - (\alpha_1/\alpha_i)\mathbf{v}_1 - \dots - (\alpha_{i-1}/\alpha_i)\mathbf{v}_{i-1} - (\alpha_{i+1}/\alpha_i)\mathbf{v}_{i+1} - \dots - (\alpha_n/\alpha_i)\mathbf{v}_n = \mathbf{v}_i \quad (= \mathbf{w}) \quad \dots \text{ and take one out}$$

By Superfluous-Vector Lemma, $\text{Span } (S \cup \{\mathbf{z}\}) = \text{Span } (S \cup \{\mathbf{z}\} - \{\mathbf{w}\})$.

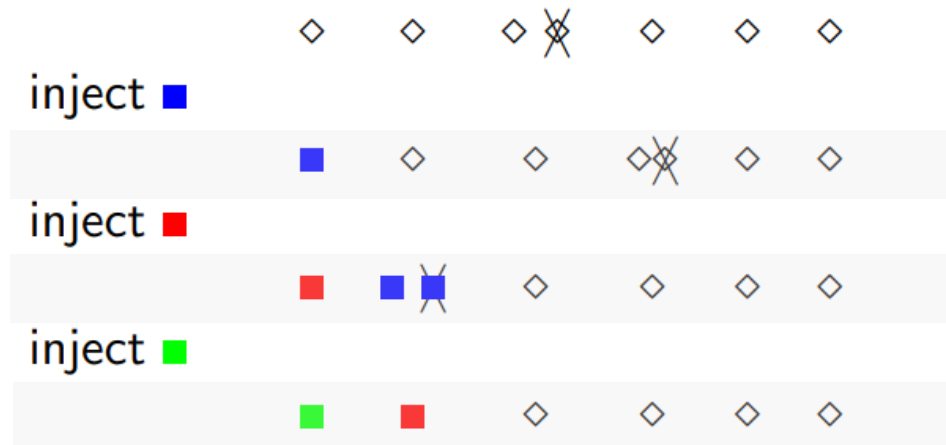
QED



Simplified Exchange Lemma (III)

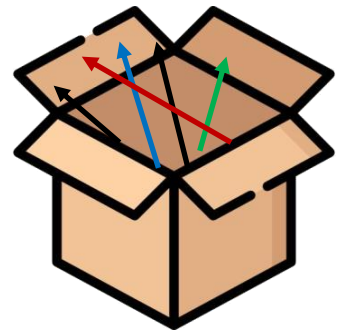
Simplified Exchange Lemma: Suppose S is a set of vectors, and $\mathbf{z}$ is a nonzero vector in $\text{Span } S$. Then there is a vector $\mathbf{w}$ in S such that $\text{Span}(S \cup \{\mathbf{z}\} - \{\mathbf{w}\}) = \text{Span } S$.

Simplified Exchange Lemma helps in transforming one generating set into another...



Trying to put squares in—when you put in one square, you might end up taking out a previously inserted square

Need a way to protect some elements from being taken out.





Exchange Lemma (I)

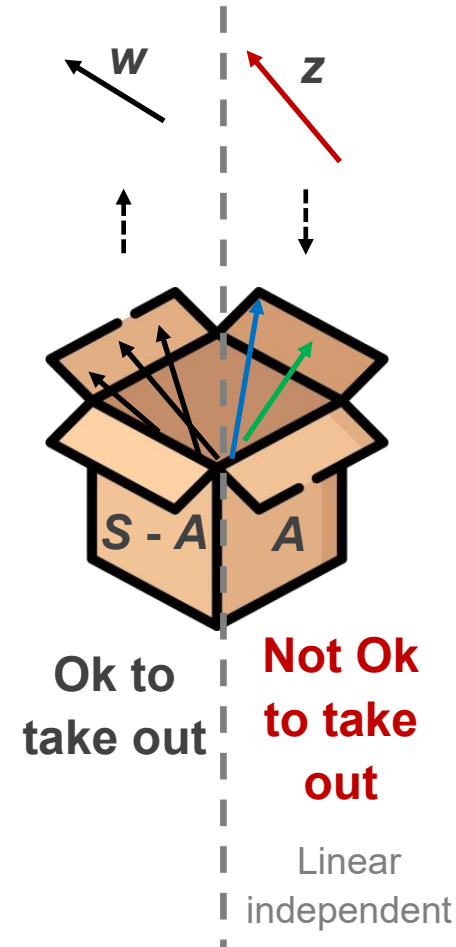
Simplified Exchange Lemma: Suppose S is a set of vectors, and $\mathbf{z}$ is a nonzero vector in $\text{Span } S$. Then there is a vector $\mathbf{w}$ in S such that $\text{Span } (S \cup \{\mathbf{z}\} - \{\mathbf{w}\}) = \text{Span } S$.

Need to enhance this lemma. Set of *protected* elements is A :

Exchange Lemma:

- ▶ Suppose S is a set of vectors and A is a subset of S .
- ▶ Suppose $\mathbf{z}$ is a vector in $\text{Span } S$ such that $A \cup \{\mathbf{z}\}$ is linearly independent.
- ▶ Then there is a vector $\mathbf{w} \in S - A$ such that $\text{Span } S = \text{Span } (S \cup \{\mathbf{z}\} - \{\mathbf{w}\})$

Now, not enough that $\mathbf{z}$ be nonzero—need A to be linearly independent.





Exchange Lemma (II)

Subset



Proof: Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}_1, \dots, \mathbf{w}_\ell\}$ and $A = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$.

Since $\mathbf{z}$ is in Span S , can write

$$\mathbf{z} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k + \beta_1 \mathbf{w}_1 + \dots + \beta_\ell \mathbf{w}_\ell$$

Adding $\mathbf{z}$
does not
change Span

Part 1

By Superfluous-Vector Lemma, $\text{Span}(S \cup \{\mathbf{z}\}) = \text{Span } S$.

If coefficients $\beta_1, \dots, \beta_\ell$ were all zero then we would have $\mathbf{z} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k$,
contradicting the linear independence of $A \cup \{\mathbf{z}\}$.

Thus one of the coefficients $\beta_1, \dots, \beta_\ell$ must be nonzero... say β_1 . Rewrite as

$$\mathbf{z} - \alpha_1 \mathbf{v}_1 - \dots - \alpha_k \mathbf{v}_k - \beta_2 \mathbf{w}_2 - \dots - \beta_\ell \mathbf{w}_\ell = \beta_1 \mathbf{w}_1$$

Vector not needed
for same Span

Part 2

Divide through by β_1 :

$$(1/\beta_1)\mathbf{z} - (\alpha_1/\beta_1)\mathbf{v}_1 - \dots - (\alpha_k/\beta_1)\mathbf{v}_k - (\beta_2/\beta_1)\mathbf{w}_2 - \dots - (\beta_\ell/\beta_1)\mathbf{w}_\ell = \mathbf{w}_1$$

By Superfluous-Vector Lemma, $\text{Span}(S \cup \{\mathbf{z}\}) = \text{Span}(S \cup \{\mathbf{z}\} - \{\mathbf{w}_1\})$. **QED**

Matrix (Basis)

Coordinates

Original Vector



$$Ax = b$$

$$Cy = b$$

Exercise: Change of Basis – 2D

You are given the following vectors and matrices:

- 1) Assume that a matrix A has a_1, a_2 as column vectors. Show whether A is a basis for $\mathbb{R}^2$.
- 2) What is the change of basis matrix B (to go from a representation via A to a representation via C)?
- 3) (Optional): Visualize both bases and the change of basis for a point with coordinates $[0.5, 0.5]^T$ according to the generators in A (as e.g., done in slide 39)

$$a_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & -1 \\ 0 & -1 \end{bmatrix} \quad C^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & -1 \end{bmatrix}$$

Solution

Exercise: Change of Basis

$$a_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & -1 \\ 0 & -1 \end{bmatrix} \quad C^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & -1 \end{bmatrix}$$

- a) We need to show that **(1) they are linearly independent**, and that **(2) the vectors are spanning** $\mathbb{R}^2$

To show that a_1 and a_2 are **linearly independent**, we need to show that the zero vector can **only** be written via a trivial linear combination (= all alpha coefficients zero) of these vectors. So,

$$\mathbf{0} = \alpha_1 a_1 + \alpha_2 a_2 \text{ only with } \alpha_1, \alpha_2, \alpha_3 = 0$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_1 + \alpha_2 \\ \alpha_2 \end{bmatrix}$$

Set of linear equations to solve

$$\begin{array}{l|l|l} \alpha_1 + \alpha_2 = 0 & \iff & \alpha_1 + \mathbf{0} = 0 \\ \alpha_2 = 0 & & \alpha_2 = \mathbf{0} \end{array} \iff \begin{array}{l|l} \alpha_1 = 0 \\ \alpha_2 = \mathbf{0} \end{array}$$

The only linear combination is the trivial one. We can conclude that a_1, a_2 are **linearly independent**. (B2)

Solution

Exercise: Change of Basis

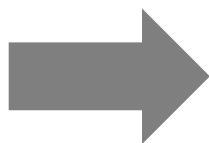
$$a_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & -1 \\ 0 & -1 \end{bmatrix} \quad C^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & -1 \end{bmatrix}$$

a) (continued)

To show that a_1 , and a_2 span $\mathbb{R}^2$, we need to show they can reach the standard generators. (As solving works as before, here the shortened solution)

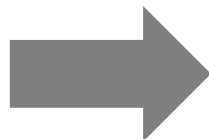
$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha_1 + \alpha_2 \\ \alpha_2 \end{bmatrix}$$



$$\alpha_1 = 1$$

$$\alpha_2 = 0$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_1 + \alpha_2 \\ \alpha_2 \end{bmatrix}$$



$$\alpha_1 = -1$$

$$\alpha_2 = 1$$

Since all standard generators for $\mathbb{R}^2$ are in the span of a_1, a_2 we can “reach” with these vectors what the standard generators can “reach”. Thus, we have demonstrated that a_1 , and a_2 also **span** $\mathbb{R}^2$. (B1)

Overall, we have shown that a_1, a_2 (1) are linearly independent and (2) span $\mathbb{R}^2$, so they form a basis for $\mathbb{R}^2$.

Solution

Exercise: Change of Basis

$$a_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & -1 \\ 0 & -1 \end{bmatrix} \quad C^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & -1 \end{bmatrix}$$

b)

y are the coordinates
via Basis A

$$Ay = b$$



x are the coordinates
via Basis C

$$C^{-1}b = x$$

$$B = C^{-1}A = \begin{bmatrix} 0.5 & -0.5 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0 \\ 0 & -1 \end{bmatrix}$$

Thus, we first need to apply A to get to the vector that is represented and then use C^{-1} to get to the new representation

(The change of basis matrix B)

Solution

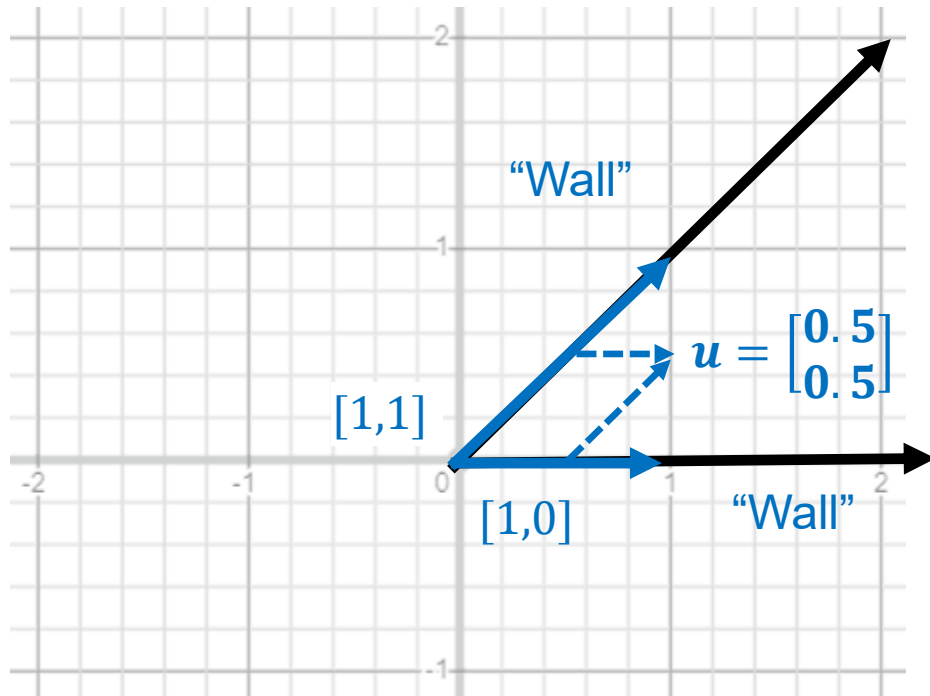
Exercise: Change of Basis

$$a_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & -1 \\ 0 & -1 \end{bmatrix} \quad C^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & -1 \end{bmatrix}$$

c) $B = \begin{bmatrix} 0.5 & 0 \\ 0 & -1 \end{bmatrix}$ $v = Bu$

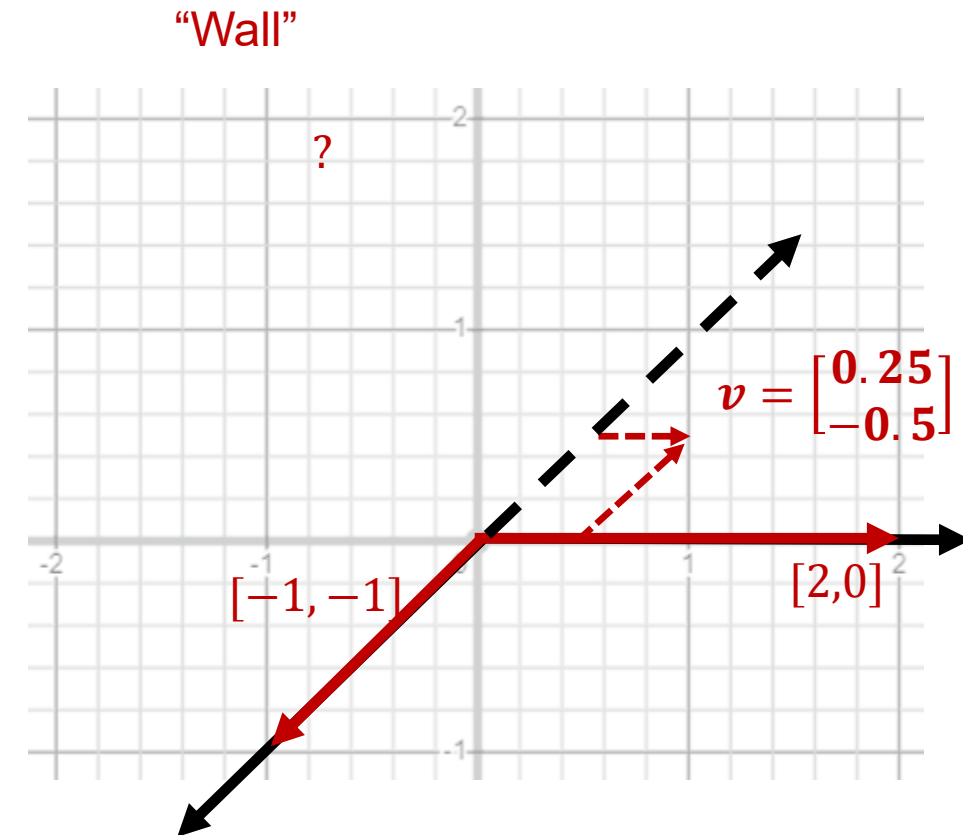
$$= \begin{bmatrix} 0.5 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.25 \\ -0.5 \end{bmatrix}$$



$b = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$

Standard Vector

$b = Au$ $C^{-1}b = v$



Matrix (Basis)

Coordinates

Original Vector



$$Ax = b$$

$$Cy = b$$

Exercise: Change of Basis – 3D

You are given the following vectors and matrices:

- 1) Assume that a matrix A has a_1, a_2, a_3 as column vectors. Show whether A is a basis for $\mathbb{R}^3$.

- 2) Assume that the coordinates for a vector b in terms of basis C are $[\frac{37}{8}, \frac{19}{4}, -\frac{7}{8}]^T$. What is the (standard) vector b we are representing in terms of the generators in C ? Show work.

- 3) What are the coordinates of the same vector b (from the previous task) in terms of A ? Show work!
- 4) What is the change of basis matrix (to go from a representation via C to a representation via A)? Show work.
- 5) Use the change of basis matrix from (4) to go from $[\frac{37}{8}, \frac{19}{4}, -\frac{7}{8}]^T$ in C to the representation in A . Compare the outcome to task (2) and (3).

$$a_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & -2 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$a_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad a_3 = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$



Solution

Exercise: Change of Basis

$$a_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad a_3 = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

a) We need to show that (1) they are **linearly independent**, and that (2) the vectors are **spanning** $\mathbb{R}^3$

To show that a_1 , a_2 , and a_3 are **linearly independent**, we need to show that the zero vector can **only** be written via a trivial linear combination (= all alpha coefficients zero) of these vectors. So,

$$\mathbf{0} = \alpha_1 a_1 + \alpha_2 a_2 + \alpha_3 a_3 \quad \text{only with} \quad \alpha_1, \alpha_2, \alpha_3 = 0$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_1 + 2\alpha_2 + 3\alpha_3 \\ \alpha_2 + 4\alpha_3 \\ \alpha_3 \end{bmatrix}$$

Set of linear equations to solve

$$\begin{array}{l|l|l|l} \begin{array}{l} \alpha_1 + 2\alpha_2 + 3\alpha_3 = 0 \\ \alpha_2 + 4\alpha_3 = 0 \\ \alpha_3 = 0 \end{array} & \longleftrightarrow & \begin{array}{l} \alpha_1 + 2\alpha_2 + 0 = 0 \\ \alpha_2 + 4(0) = 0 \\ \alpha_3 = 0 \end{array} & \longleftrightarrow & \begin{array}{l} \alpha_1 + 2(0) = 0 \\ \alpha_2 = 0 \\ \alpha_3 = 0 \end{array} & \longleftrightarrow & \begin{array}{l} \alpha_1 = 0 \\ \alpha_2 = 0 \\ \alpha_3 = 0 \end{array} \end{array}$$

The only linear combination is the trivial one. We can conclude that a_1, a_2, a_3 are **linearly independent**.



Solution

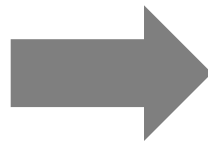
Exercise: Change of Basis

$$a_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad a_3 = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

a)

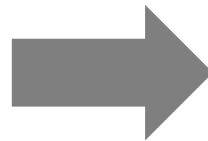
To show that a_1 , a_2 , and a_3 span $\mathbb{R}^3$, we need to show they can reach the standard generators. (As solving works as before, here the shortened solution)

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha_1 + 2\alpha_2 + 3\alpha_3 \\ \alpha_2 + 4\alpha_3 \\ \alpha_3 \end{bmatrix}$$



$$\begin{aligned} \alpha_1 &= 1 \\ \alpha_2 &= 0 \\ \alpha_3 &= 0 \end{aligned}$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha_1 + 2\alpha_2 + 3\alpha_3 \\ \alpha_2 + 4\alpha_3 \\ \alpha_3 \end{bmatrix}$$



$$\begin{aligned} \alpha_1 &= -2 \\ \alpha_2 &= 1 \\ \alpha_3 &= 0 \end{aligned}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_1 + 2\alpha_2 + 3\alpha_3 \\ \alpha_2 + 4\alpha_3 \\ \alpha_3 \end{bmatrix}$$



$$\begin{aligned} \alpha_1 &= 8 \\ \alpha_2 &= -4 \\ \alpha_3 &= 1 \end{aligned}$$

Since all standard generators for $\mathbb{R}^3$ are in the span of a_1, a_2, a_3 , we can “reach” with these vectors what the standard generators can “reach”. Thus, we have demonstrated that a_1 , a_2 , and a_3 also **span** $\mathbb{R}^3$.

Overall, we have shown that a_1, a_2, a_3 (1) are linearly independent and (2) span $\mathbb{R}^3$, so they form a basis for $\mathbb{R}^3$.

Solution

Exercise: Change of Basis

b) $Cy = b$

$$b = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix} \times \begin{bmatrix} \frac{37}{8} \\ \frac{19}{4} \\ -\frac{7}{8} \end{bmatrix} = \begin{bmatrix} 14 \\ 18 \\ 3 \end{bmatrix}$$

c) $Ax = b$

$$A^{-1}Ax = A^{-1}b$$

$$x = A^{-1}b$$

$$x = \begin{bmatrix} 1 & -2 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 14 \\ 18 \\ 3 \end{bmatrix} = \begin{bmatrix} -7 \\ 6 \\ 3 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} 1 & -2 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$y = \left[\frac{37}{8}, \frac{19}{4}, -\frac{7}{8} \right]$$



Thus, the vector that is represented via y terms of the basis C is $x = \begin{bmatrix} 14 \\ 18 \\ 3 \end{bmatrix}$.

Thus, the coordinate representation of b in terms of the basis A is $x = \begin{bmatrix} -7 \\ 6 \\ 3 \end{bmatrix}$.



Solution

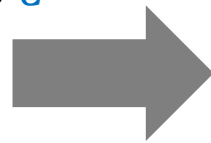
Exercise: Change of Basis

$$y = \left[\frac{37}{8}, \frac{19}{4}, -\frac{7}{8}\right] \quad C = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

d)

y are the coordinates via Basis C x are the coordinates via Basis A

$$Cy = b$$



$$A^{-1}b = x$$

Thus, we first need to apply C to get to the vector that is represented and then use A^{-1} to get to the new representation

$$A^{-1}C = \begin{bmatrix} 1 & -2 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 8 \\ 1 & -1 & -7 \\ 0 & 1 & 2 \end{bmatrix}$$

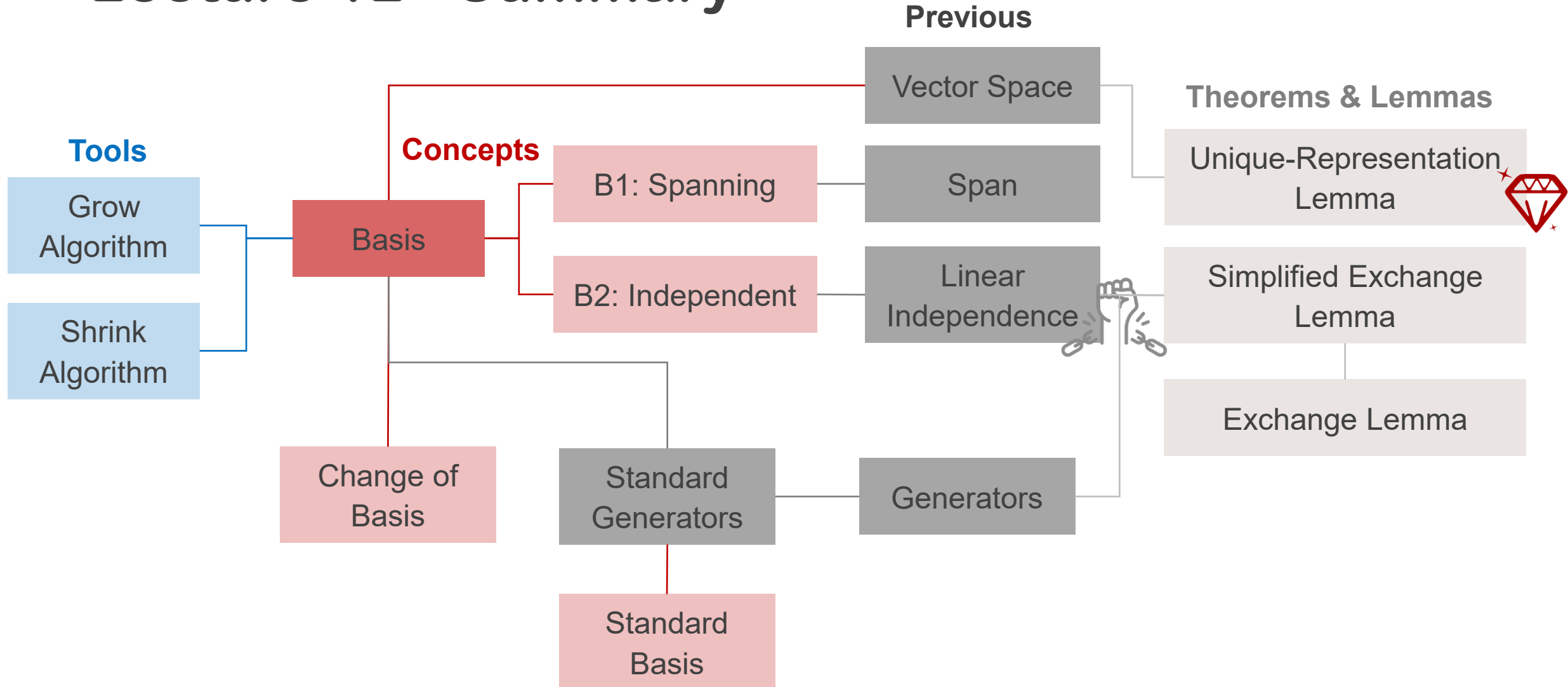
e)

$$(A^{-1}C)y = \begin{bmatrix} 0 & 0 & 8 \\ 1 & -1 & -7 \\ 0 & 1 & 2 \end{bmatrix} \times \begin{bmatrix} \frac{37}{8} \\ \frac{19}{4} \\ -\frac{7}{8} \end{bmatrix} = \begin{bmatrix} -7 \\ 6 \\ 3 \end{bmatrix}$$

Unsurprisingly, the results reflect task (b) and (c) and show that the change of basis matrix does in one step, what we did in (b) and (c) in two separate steps.



Lecture 12 - Summary





Some Key Take-aways (Lecture 12)

A **basis** for a vector space is a linearly independent set of generators for that vector space. A set of vectors is a basis for a vector space if it satisfies **two properties** – they span the vector space (B1) are linearly independent (B2)

Linear
Independence is
essential

The standard generators form a **standard basis**

The **Unique-Representation Lemma** states that for a basis for a vector space there is only one representation of each vector in that vector space using the basis vectors

Given two bases, there exists a matrix such that multiplication with that matrix converts one coordinate representation to another (**change of basis**)

The **Simplified Exchange Lemma** states that for a set of vectors, a vector in that set that is in the span of the other vectors can be removed and another one added, without changing the span. The **Exchange Lemma** additionally adds the requirement that vectors retained and added are linearly independent

Transforming one
set of generators to
another



Thank You For Your Attention!



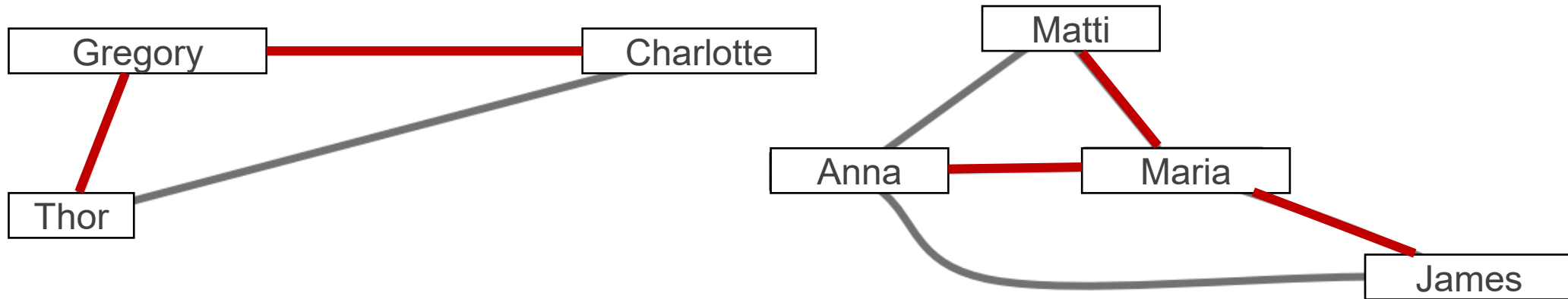
**Accomplishment unlocked:
Lecture 12 Completed**



APPENDIX



Basis - Graph Examples

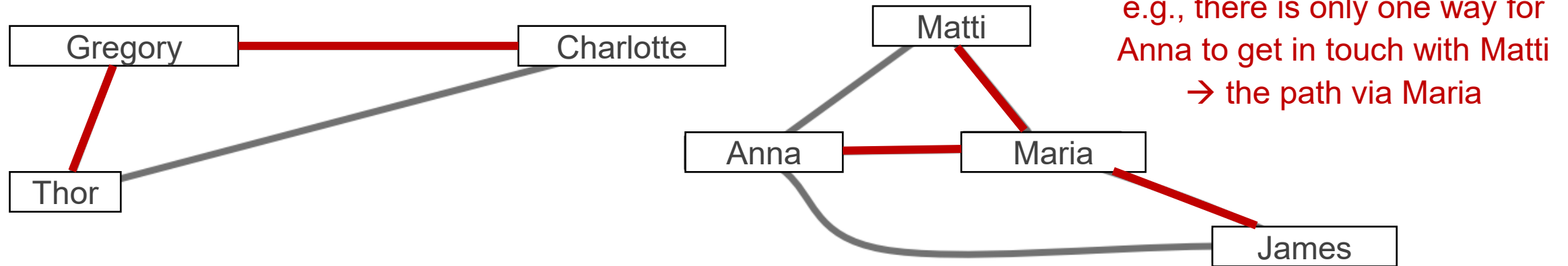


One kind of basis in a graph G : a set S of edges forming a spanning forest.

- B1 ▶ *Spanning*: for each edge xy in G , there is an x -to- y path consisting of edges of S .
- B2 ▶ *Independent*: no cycle consisting of edges of S



Unique Representation via Basis - Graphs



A basis for a graph is a spanning forest.

Unique Representation shows that, for each edge xy in the graph,

- ▶ there is an x -to- y path in the spanning forest, and 
- ▶ there is only one such path. 

Remember
(some slides ago)

Definition: A set of edges of G is a *forest* if the set includes no cycles.



Every Vector Space has a Basis

We would like to prove that every vector space $\mathcal{V}$ has a basis.

The Grow algorithm and the Shrink algorithm each provides a way to prove this, but we are not there yet:

-
- ▶ The Grow-Algorithm Corollary implies that, if the Grow algorithm terminates, the set of vectors it has selected is a basis for the vector space $\mathcal{V}$.

However, we have *not yet shown that it always terminates!*



-
- ▶ The Shrink-Algorithm Corollary implies that, if we can run the Shrink algorithm starting with a finite set of vectors that spans $\mathcal{V}$, upon termination it will have selected a basis for $\mathcal{V}$.

However, we have *not yet shown that every vector space $\mathcal{V}$ is spanned by some finite set of vectors!*





Basis - Computational Problems

Two natural ways
to specify a vector
space $\mathcal{V}$

1. Specifying generators for $\mathcal{V}$.
2. Specifying a homogeneous linear system whose solution set is $\mathcal{V}$.

Two fundamental
computational
problems

Computational Problem: *Finding a basis of the vector space spanned by given vectors*

- ▶ *input:* a list $[\mathbf{v}_1, \dots, \mathbf{v}_n]$ of vectors
- ▶ *output:* a list of vectors that form a basis for $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

Computational Problem: *Finding a basis of the solution set of a homogeneous linear system*

- ▶ *input:* a list $[\mathbf{a}_1, \dots, \mathbf{a}_n]$ of vectors
- ▶ *output:* a list of vectors that form a basis for the set of solutions to the system $\mathbf{a}_1 \cdot \mathbf{x} = 0, \dots, \mathbf{a}_n \cdot \mathbf{x} = 0$



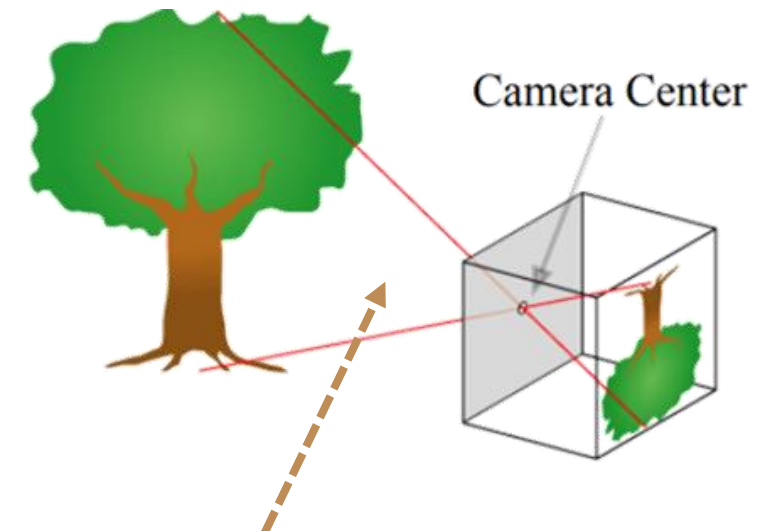
Simplified Camera Model (I)

As application of change of basis, we show how to synthesize a camera view from a set of points in three dimensions, taking into account perspective.

We can also e.g., remove perspective from a real image

- ▶ There is a point called the camera center.
- ▶ There is an image sensor array in the back of the camera.
- ▶ Photons bounce off objects in the scene and travel through the camera center to the image sensor array.
- ▶ A photon from the scene only reaches the image sensor array if it travels in a straight line through the camera center.
- ▶ The image ends up being inverted.

Simplified model of a camera:



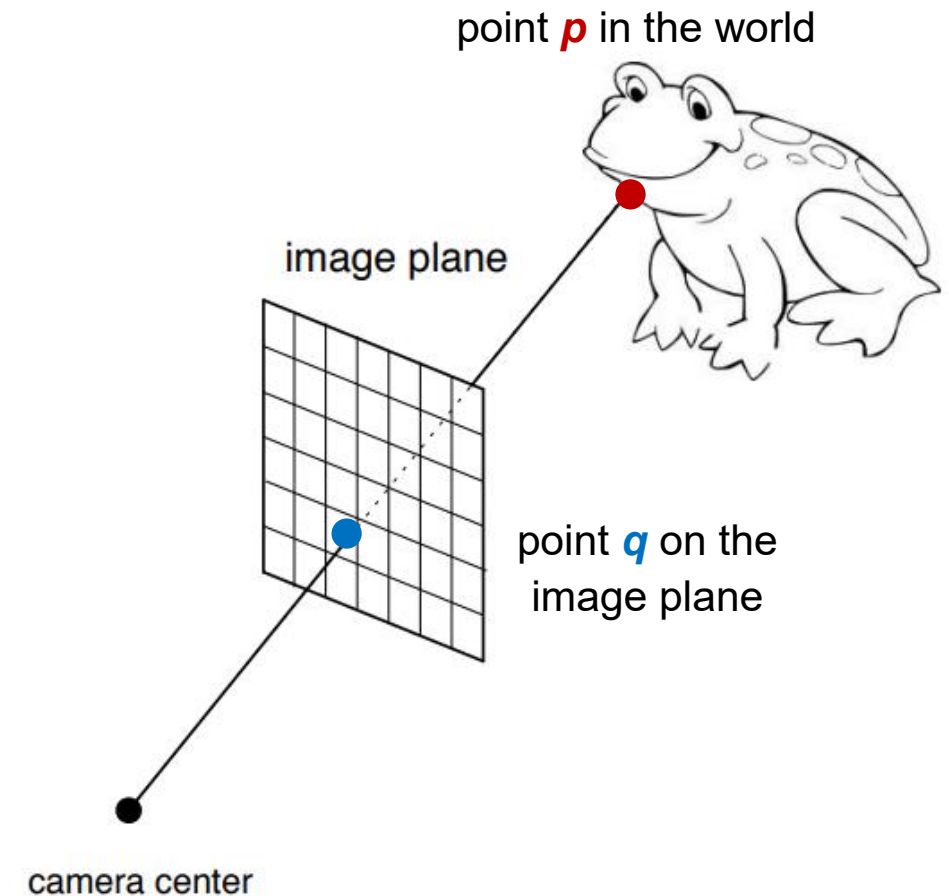
Next: Put image sensor here



Simplified Camera Model (II)

Even simpler model to avoid the inversion:

- ▶ The image sensor array is between the camera center and the scene.
- ▶ The image sensor array is located in a plane, called the *image plane*.
- ▶ A photon from the scene is detected by the sensor array only if it is traveling in a straight line towards the camera center.
- ▶ The sensor element that detects the photon is the one intersected by this line.
- ▶ Need a function that maps from point p in world to corresponding point q in image plane

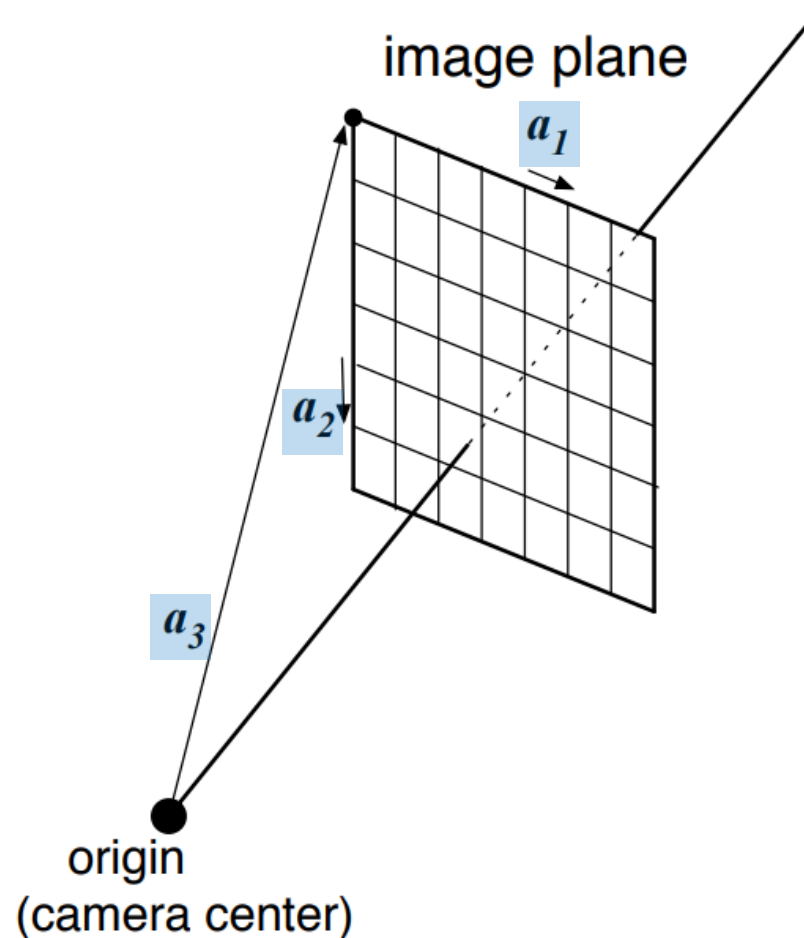




Simplified Camera Model (III)

Camera-oriented basis helps in mapping from world points to image-plane points:

- ▶ The origin is defined to be the camera center.
(That's why we translated the wire-frame cube.)
- ▶ The first vector $\mathbf{a}_1$ goes horizontally from the top-left corner of a sensor element to the top-right corner.
- ▶ The second vector $\mathbf{a}_2$ goes vertically from the top-left corner of a sensor element to the bottom-left corner.
- ▶ The third vector $\mathbf{a}_3$ goes from the origin (the camera center) to the top-left corner of sensor element (0,0).



Next: Looking
“only” from here





World Point to Camera Point

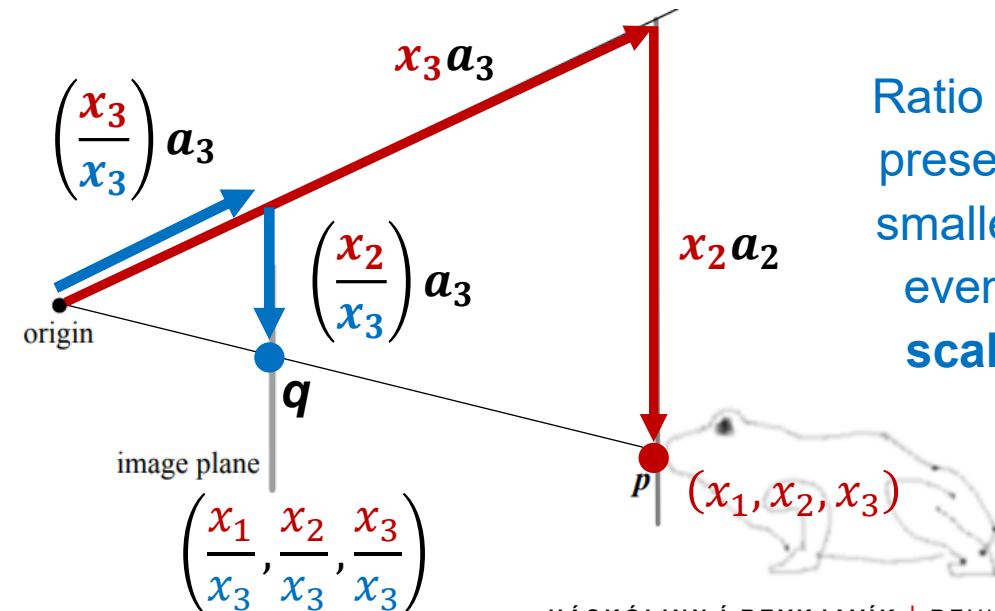
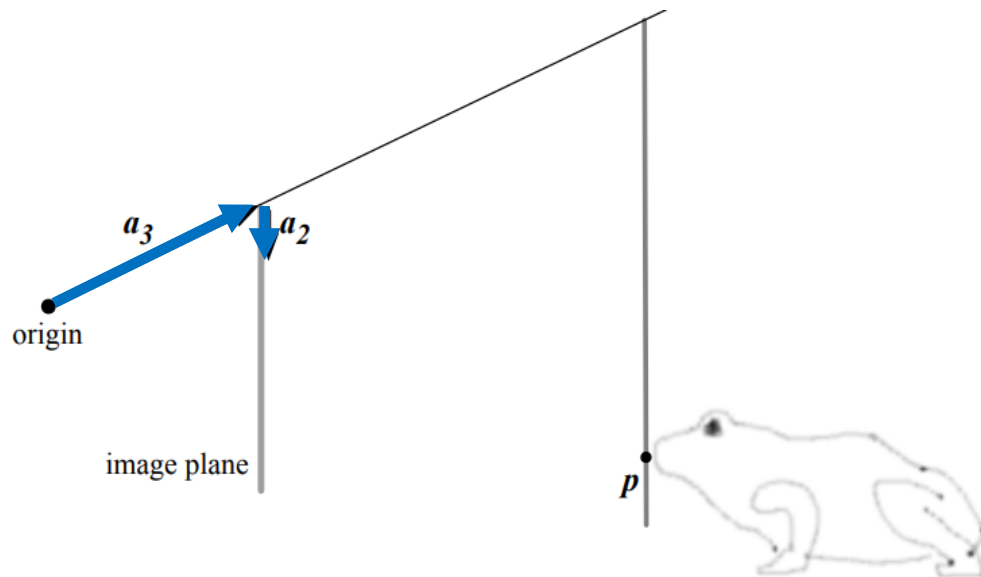
(a_1 is currently not shown and is towards the viewer of the slides)

Side view (we see only the edge of the image plane)

- ▶ Have a point $\mathbf{p}$ in the world
- ▶ Express it in terms of $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$
- ▶ Consider corresponding point $\mathbf{q}$ in image plane.
- ▶ Similar triangles $\Rightarrow$ coordinates of $\mathbf{q}$

Summary: Given coordinate representation (x_1, x_2, x_3) in terms of $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$,

coordinate representation of corresponding point in image plane is $(x_1/x_3, x_2/x_3, x_3/x_3)$.



Ratio of sides are preserved (in the smaller triangle – even for x_1) $\rightarrow$ **scaling down**

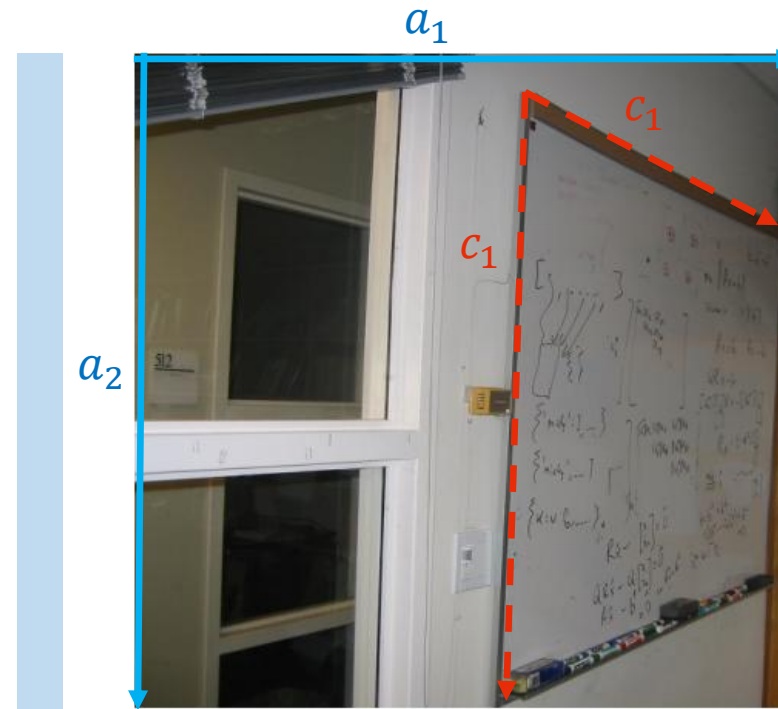


Removing Perspective

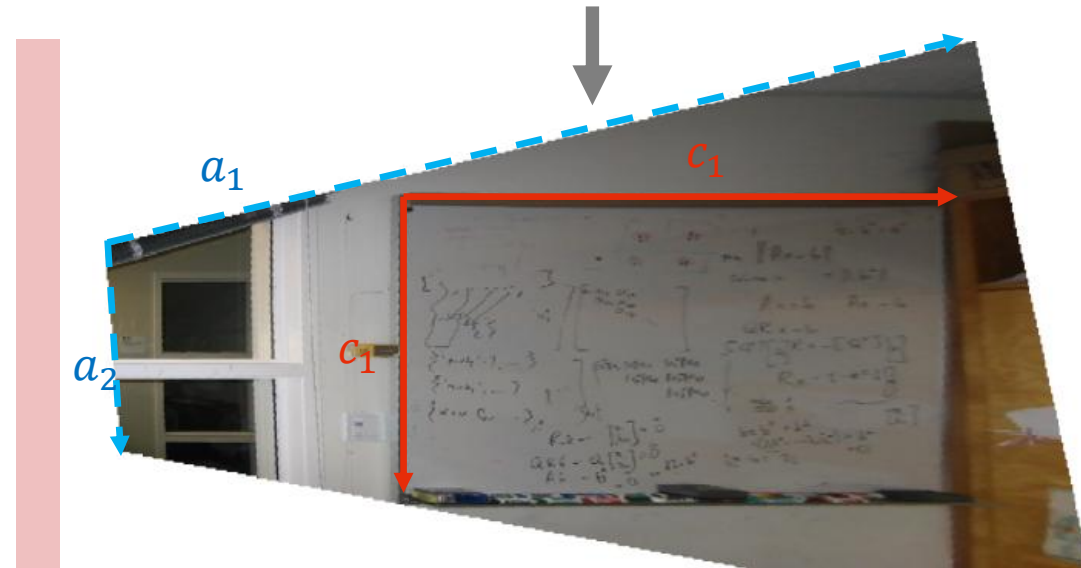
Given an image of a whiteboard, taken from an angle...

synthesize an image
from straight ahead
with no perspective

We use the **camera basis**
and the **whiteboard basis**
to do that



**Camera
Perspective**



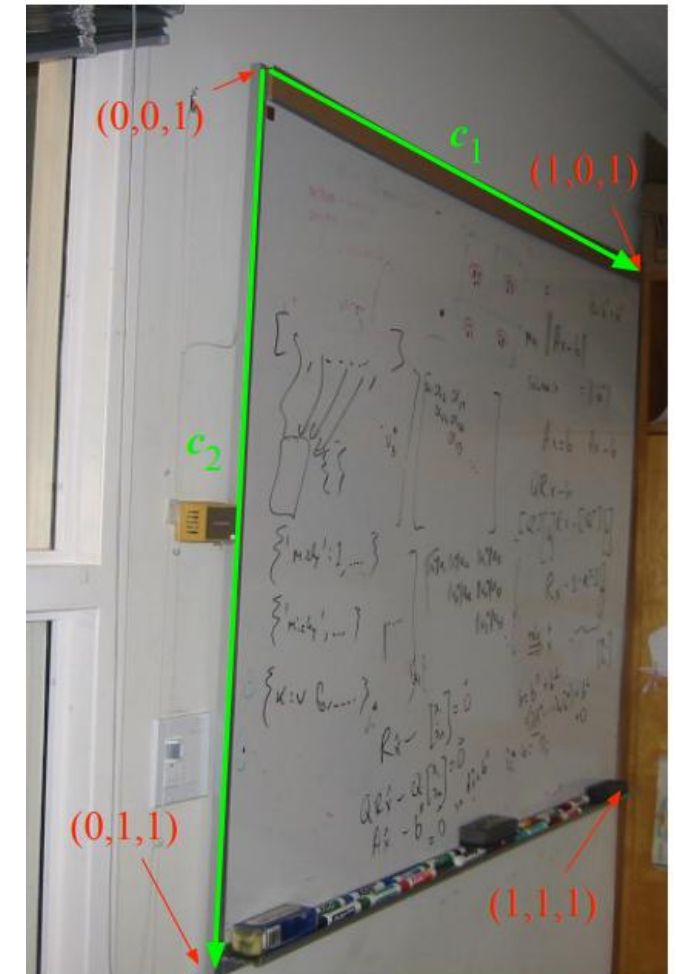
**Whiteboard
Perspective**



Converting from one Basis to another (I)

In addition, we define a *whiteboard basis* $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$

- ▶ The origin is the camera center.
- ▶ The first vector $\mathbf{c}_1$ goes horizontally from the top-left corner of whiteboard to top-right corner.
- ▶ The second vector $\mathbf{c}_2$ goes vertically from the top-left corner of whiteboard to the bottom-left corner.
- ▶ The third vector $\mathbf{c}_3$ goes from the origin (the camera center) to the top-left corner of whiteboard.





Converting from one Basis to another (II)

Start with a point $\mathbf{p}$ written in terms of in camera coordinates

$$\text{Original Vector (Actual Point)} \quad \mathbf{p} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

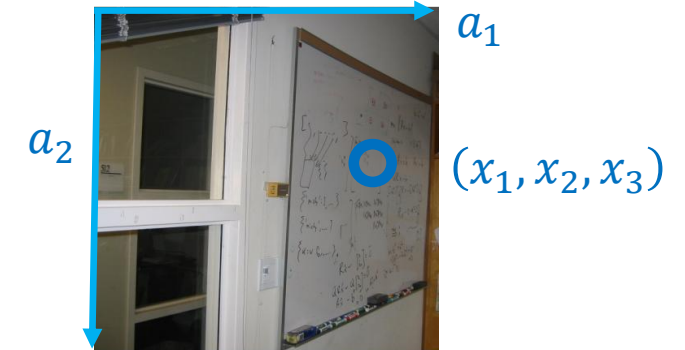
We write the same point $\mathbf{p}$ in the whiteboard coordinate system as

$$\text{Original Vector (Actual Point)} \quad \mathbf{p} = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

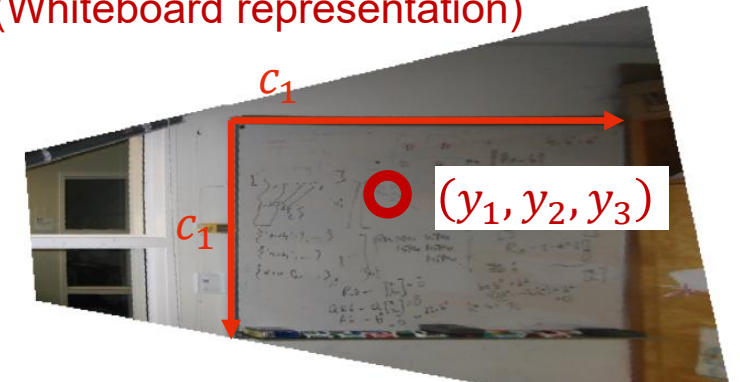
Combining the two equations, we obtain

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

(Camera representation)



(Whiteboard representation)





Converting from one Basis to another (III)

Let A and C be the two matrices. As before, C has an inverse C^{-1} .

Multiplying equation on the left by C^{-1} , we obtain

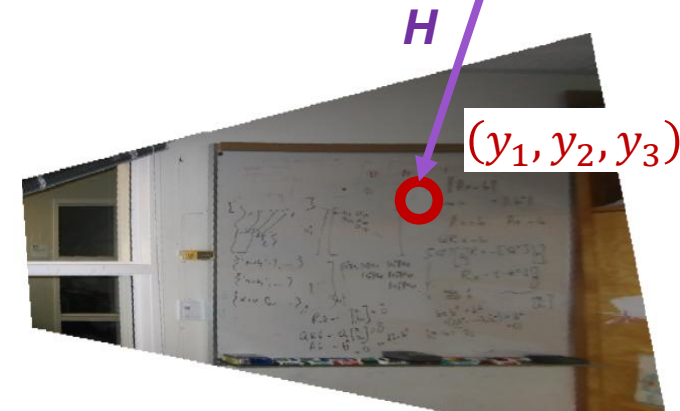
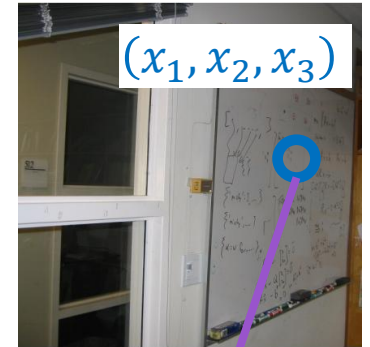
$$\begin{bmatrix} C^{-1} \end{bmatrix} \begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} C^{-1} \end{bmatrix} \begin{bmatrix} C \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

Since C^{-1} and C cancel out, we obtain

$$\begin{bmatrix} C^{-1} \end{bmatrix} \begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

We have shown that there is a matrix H (namely $H = C^{-1}A$) such that

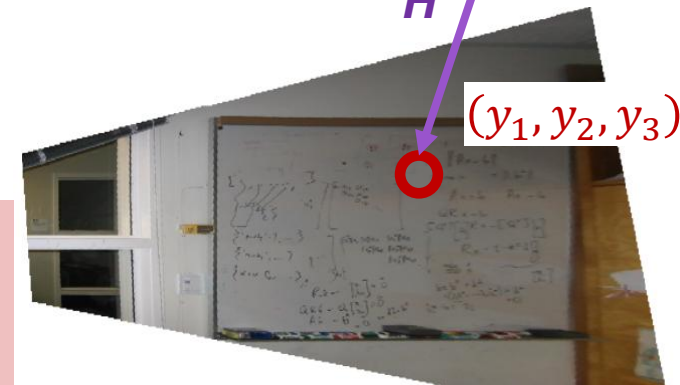
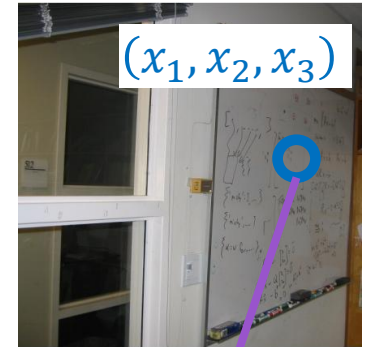
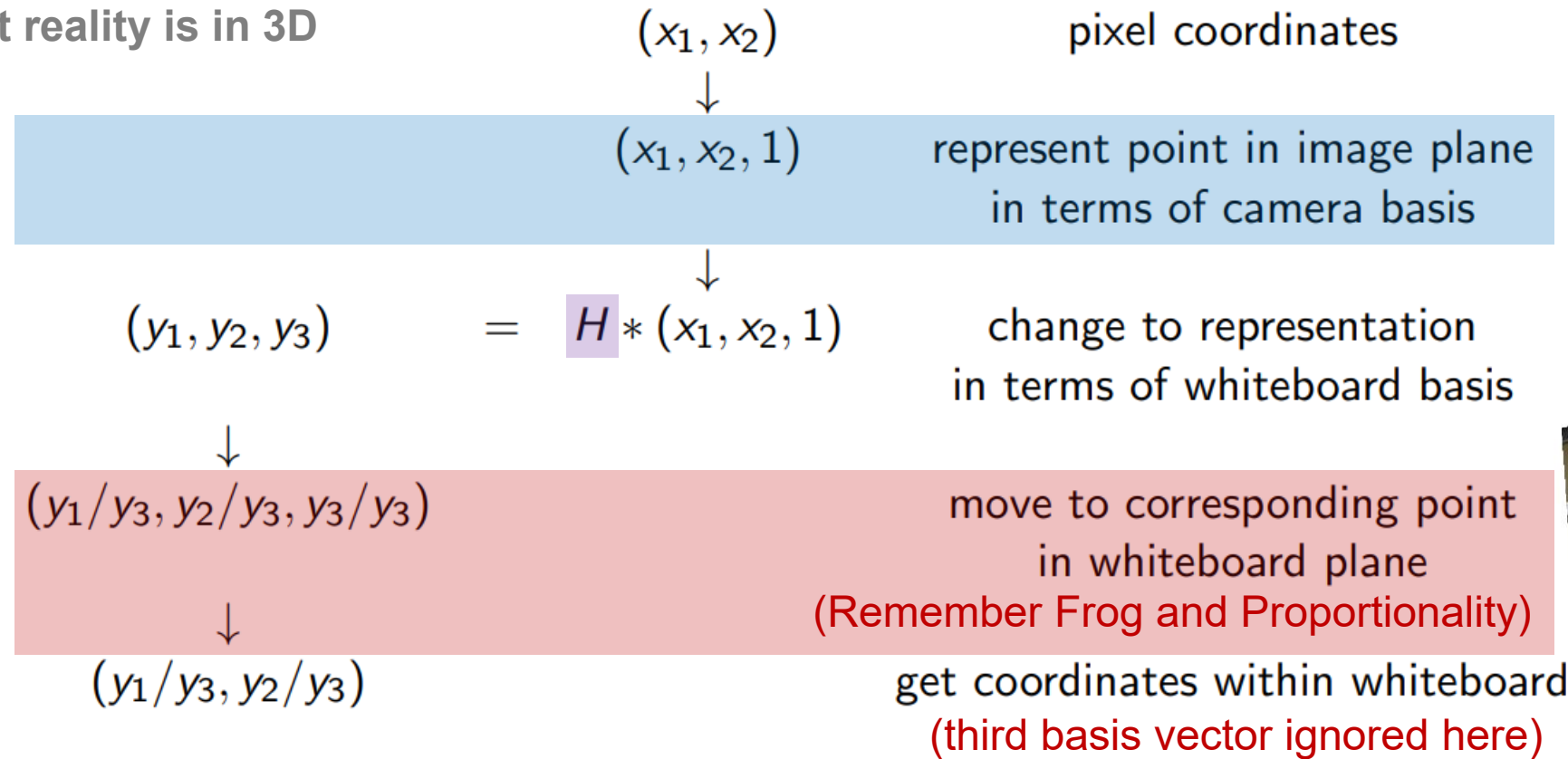
$$\begin{bmatrix} H \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$





Converting from one Basis to another (IV)

The picture is in 2D
but reality is in 3D





Converting from one Basis to another (V)

Write $H = \begin{bmatrix} h_{y_1, x_1} & h_{y_1, x_2} & h_{y_1, x_3} \\ h_{y_2, x_1} & h_{y_2, x_2} & h_{y_2, x_3} \\ h_{y_3, x_1} & h_{y_3, x_2} & h_{y_3, x_3} \end{bmatrix}$



(we don't know the bases)

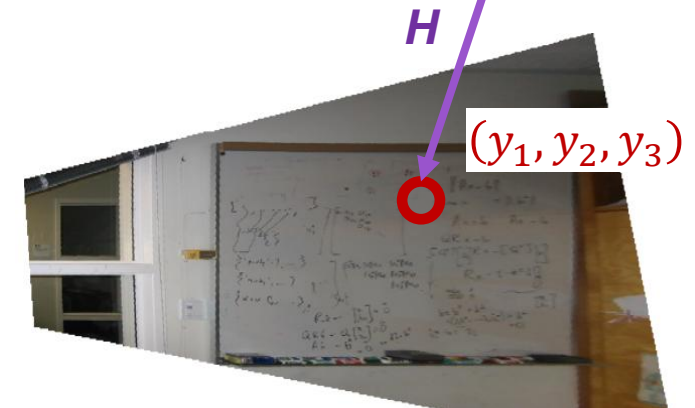
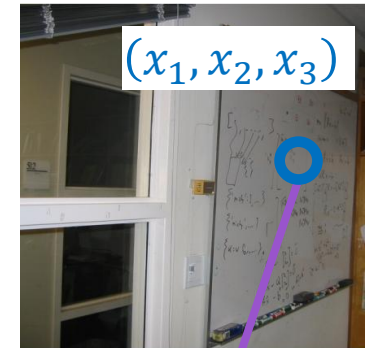
The h_{ij} 's are the **unknowns**.

To derive equations, let $\mathbf{p}$ be some point on the whiteboard, and let $\mathbf{q}$ be the corresponding point on the image plane. Let $(x_1, x_2, 1)$ be the **camera coordinates** of $\mathbf{q}$, and let (y_1, y_2, y_3) be the **whiteboard coordinates** of $\mathbf{q}$. We have

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} h_{y_1, x_1} & h_{y_1, x_2} & h_{y_1, x_3} \\ h_{y_2, x_1} & h_{y_2, x_2} & h_{y_2, x_3} \\ h_{y_3, x_1} & h_{y_3, x_2} & h_{y_3, x_3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

Multiplying out, we obtain

$$\begin{aligned} y_1 &= h_{y_1, x_1} x_1 + h_{y_1, x_2} x_2 + h_{y_1, x_3} \\ y_2 &= h_{y_2, x_1} x_1 + h_{y_2, x_2} x_2 + h_{y_2, x_3} \\ y_3 &= h_{y_3, x_1} x_1 + h_{y_3, x_2} x_2 + h_{y_3, x_3} \end{aligned}$$





Almost computing $H(I)$

$$y_1 = h_{y_1,x_1}x_1 + h_{y_1,x_2}x_2 + h_{y_1,x_3}$$

$$y_2 = h_{y_2,x_1}x_1 + h_{y_2,x_2}x_2 + h_{y_2,x_3}$$

$$y_3 = h_{y_3,x_1}x_1 + h_{y_3,x_2}x_2 + h_{y_3,x_3}$$

(no equation has the same entries of H ...)

Whiteboard coordinates of the original point $\mathbf{p}$ are $(y_1/y_3, y_2/y_3, 1)$. Define

$$w_1 = y_1/y_3$$

$$w_2 = y_2/y_3$$

(think about frog picture)

so the whiteboard coordinates of $\mathbf{p}$ are $(w_1, w_2, 1)$. Multiplying through by y_3 , we obtain

$$w_1 y_3 = y_1$$

$$w_2 y_3 = y_2$$

Substituting our expressions for y_1, y_2, y_3 , we obtain

$$w_1(h_{y_3,x_1}x_1 + h_{y_3,x_2}x_2 + h_{y_3,x_3}) = h_{y_1,x_1}x_1 + h_{y_1,x_2}x_2 + h_{y_1,x_3}$$

$$w_2(h_{y_3,x_1}x_1 + h_{y_3,x_2}x_2 + h_{y_3,x_3}) = h_{y_2,x_1}x_1 + h_{y_2,x_2}x_2 + h_{y_2,x_3}$$





Almost computing $H(II)$

$$\begin{aligned} w_1(h_{y_3,x_1}x_1 + h_{y_3,x_2}x_2 + h_{y_3,x_3}) &= h_{y_1,x_1}x_1 + h_{y_1,x_2}x_2 + h_{y_1,x_3} \\ w_2(h_{y_3,x_1}x_1 + h_{y_3,x_2}x_2 + h_{y_3,x_3}) &= h_{y_2,x_1}x_1 + h_{y_2,x_2}x_2 + h_{y_2,x_3} \end{aligned}$$

(repeated from
end of last slide)

Multiplying through and moving everything to the same side, we obtain

$$\begin{aligned} (w_1x_1)h_{y_3,x_1} + (w_1x_2)h_{y_3,x_2} + w_1h_{y_3,x_3} - x_1h_{y_1,x_1} - x_2h_{y_1,x_2} - 1h_{y_1,x_3} &= 0 \\ (w_2x_1)h_{y_3,x_1} + (w_2x_2)h_{y_3,x_2} + w_2h_{y_3,x_3} - x_1h_{y_2,x_1} - x_2h_{y_2,x_2} - 1h_{y_2,x_3} &= 0 \end{aligned}$$

2 equations, 9
unknowns

Thus we get two linear equations in the unknowns. The coefficients are expressed in terms of x_1, x_2, w_1, w_2 . [Corners of the whiteboard]

For four points, get eight equations. Need one more...
(e.g., corners of the whiteboard)

[More known points would not help]

8 equations $\longleftrightarrow$ 9 unknowns





Almost computing H (III)

We can't pin down H precisely.

This corresponds to the fact that we cannot recover the scale of the picture (a tiny building that is nearby looks just like a huge building that is far away).

Fortunately, we don't need the true H .

[Here: the whiteboard could be small and close OR large and far]

As long as the H we compute is a scalar multiple of the true H , things will work out.

To arbitrarily select a scale, we add the equation $h_{y_1, x_1} = 1$.

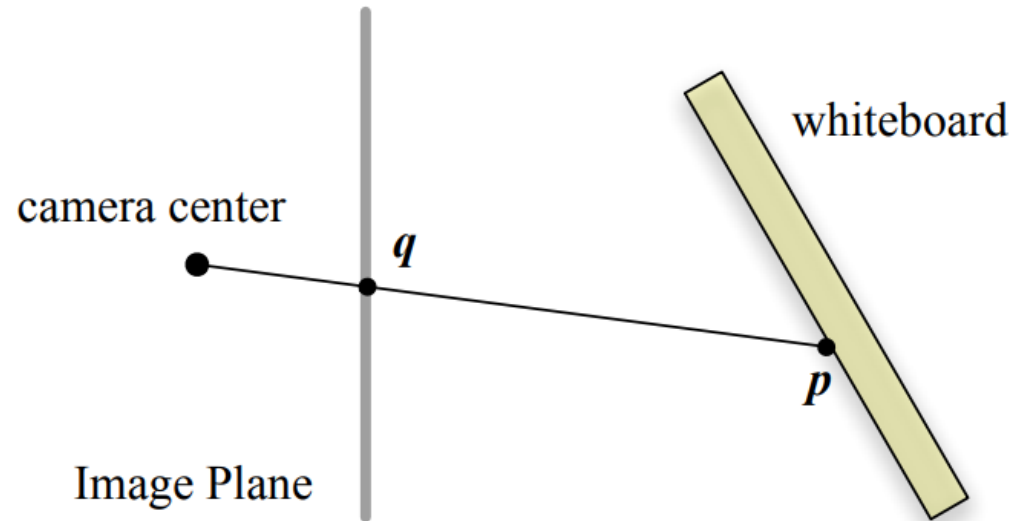
9 equations  $\longleftrightarrow$ 9 unknowns





Once we have H

1. For each point $\mathbf{q}$ in the representation of the image, we have the camera coordinates $(x_1, x_2, 1)$ of $\mathbf{q}$. We multiply by H to obtain the whiteboard coordinates (y_1, y_2, y_3) of the same point $\mathbf{q}$.
2. Recall the situation as viewed from above:

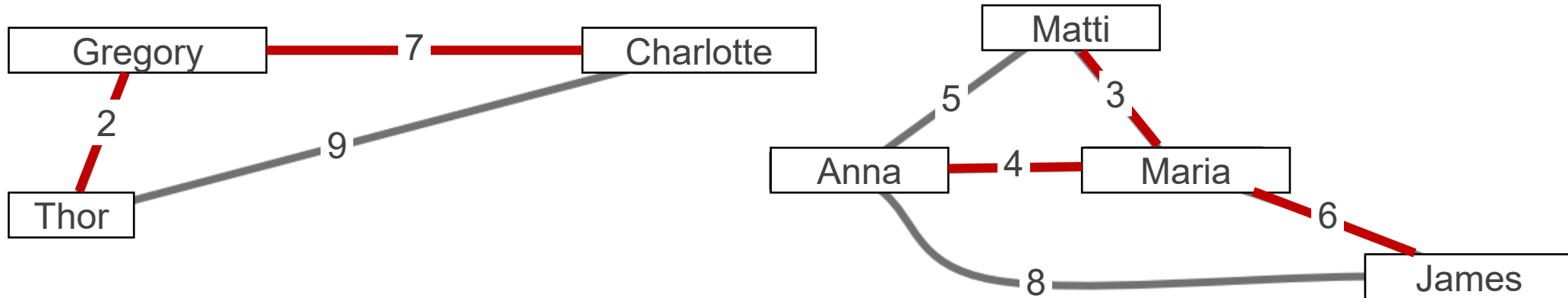


The whiteboard coordinates of the corresponding point $\mathbf{p}$ on the whiteboard are $(y_1/y_3, y_2/y_3, 1)$. Use this formula to compute these coordinates.

3. Display the updated points with the same color matrix



Correctness of the Grow Algorithm for MST (I)



```
def GROW( $G$ )
   $F := \emptyset$ 
  consider the edges in increasing order
  for each edge  $e$ :
    if  $e$ 's endpoints are not yet connected
      add  $e$  to  $F$ .
```

We will show that this greedy algorithm chooses the minimum-weight spanning forest.

(Assume all weights are distinct.)

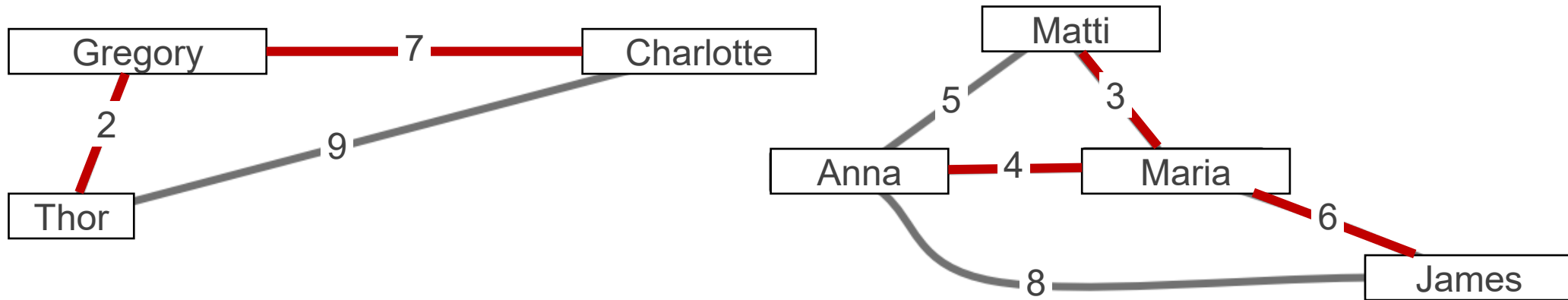
Let F = forest found by algorithm.

Let F^* = truly minimum-weight spanning forest.

Goal: show that $F = F^*$ Assume for a contradiction that they are different.



Correctness of the Grow Algorithm for MST (II)



Our MST Optimal MST

Assume for a contradiction that F and F^* are different.

Let $e_1, e_2, \dots, e_m$ be the edges of G in increasing order.

Let e_k be the minimum-weight edge on which F and F^* disagree.

Let A be the set of edges before e_k that are in both F and F^* . What they agree on (up to e_k)



Correctness of the Grow Algorithm for MST (III)



Since at least one of the forests includes all of A and also e_k , we know $A \cup \{e_k\}$ has no cycles (is linearly independent).

Consider the moment when the Grow algorithm considers e_k . So far, the algorithm has chosen the edges in A , and e_k does not form a cycle with edges in A , so the algorithm must also choose e_k .

Since F and F^* differ on e_k , we infer that e_k is *not* in F^* .

Now we use the **Exchange Lemma**.

- ▶ A is a subset of F^* .
- ▶ $A \cup \{e_k\}$ is linearly independent.
- ▶ Therefore there is an edge e_n in $F^* - A$ such that

$$\text{Span}(F^* \cup \{e_k\} - \{e_n\}) = \text{Span } F^*$$

We may be able keep the Span by exchanging e_k for e_n , but that would be at a higher “cost”

That is, $F^* \cup \{e_k\} - \{e_n\}$ is also spanning.

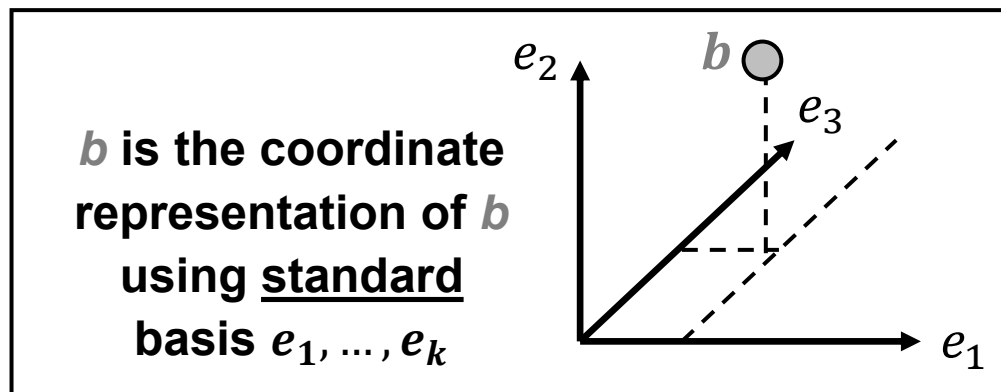
But e_k is cheaper than e_n so F^* is not min-weight solution. **Contradiction.** QED.



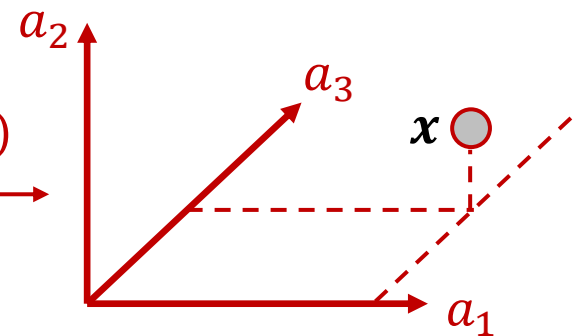
Change of Basis (VII)

Remark: Converting between vector itself and its coordinate representation is a special case:

- Think of the vector itself as coordinate representation with respect to standard basis. Using (standard generators $e_1, \dots, e_k$)



$f^{-1}(b)$



x is the coordinate representation of b using basis $a_1, \dots, a_n$



Change of Basis (XIII)

Example: To map from coordinate representation with respect to $[1, 2, 3], [2, 1, 0], [0, 1, 4]$ to coordinate representation with respect to $[2, 0, 1], [0, 1, -1], [1, 2, 0]$

$$f^{-1}(g(y))$$

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ 3 & 0 & 4 \end{bmatrix}$$

which is

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{4}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ 3 & 0 & 4 \end{bmatrix}$$

which is

$$\begin{bmatrix} -1 & 1 & -\frac{5}{3} \\ -4 & 1 & -\frac{17}{3} \\ 3 & 9 & \frac{10}{3} \end{bmatrix} = \mathbf{B}$$

Matrix $\mathbf{B}$ is the matrix to implement $f^{-1}(g(y))$ by multiplying it with y